# Predicting India's Weighted Average Call Money Rate

## An End-to-End Data Science Project on RBI NDAP and Market Data, with a Live Dashboard, Counterfactual Simulator, and AI-Powered Data Agent

**Aamer Jalan and Arnav Goyal**

Ashoka University, Data Science and Management

April 26, 2026

**Summary.** This report documents an end-to-end data science investigation into the Weighted Average Call Money Rate (WACMR), the operational target of the Reserve Bank of India's (RBI) Liquidity Adjustment Facility and the cleanest market-revealed signal of overnight rupee liquidity. Ten public datasets were ingested, eight from the NDAP open-data API and two from Yahoo Finance, then aligned onto a canonical Friday weekly grid spanning February 2014 to July 2024. The resulting panel of 545 observations across 117 features was stored in SQLite and analysed through three complementary lenses, namely unsupervised regime discovery via PCA and K-Means, supervised forecasting via walk-forward XGBoost with SHAP interpretability, and a domain-curated NLP corpus of seventy-five RBI policy events. The findings are deployed as a live, publicly accessible web application that includes a policy counterfactual simulator and a Gemini-powered data agent.

**Project highlights.**

| | | | | |
|---|---|---|---|---|
| **545** weeks of weekly macro data | **117** engineered features | **10** data sources | **70.9%** directional accuracy | **75** curated policy events |

| | |
|---|---|
| **Data scope** | February 2014 to July 2024, weekly Friday-aligned panel, 545 rows by 119 columns in SQLite, target variable `target_wacmr` (RBI `rates_I7496_26`). |
| **Analytical methods** | PCA with 90.4% variance retention (14 components), K-Means with K=2 (silhouette 0.4238), walk-forward expanding-window XGBoost over 389 test steps, SHAP TreeExplainer, and a manually curated NLP event corpus. |
| **Headline result** | Baseline XGBoost achieves RMSE 0.1019 (about ten basis points) and 70.9% directional accuracy. None of 28 equity and forex features rank in the top 15 SHAP drivers, confirming that WACMR is purely corridor-bound at the weekly horizon. |
| **Live system** | Next.js front-end on Vercel, FastAPI backend on Render, Gemini 2.5 Flash function-calling agent over the dataset, and a real-time policy counterfactual slider with cached inference (26 seconds compressed to roughly 50 ms). |

**Public deployment.**
**Dashboard and blog:** https://dsm-project-phi.vercel.app
**API and agent:** https://wacmr-api.onrender.com
**Source repository:** https://github.com/AJ112103/DSM_project

*All figures cited in this report are reproducible from the code committed to the repository. Pipeline stages are numbered `stage1_*` through `stage6_*` and are idempotent.*

# Contents

**1 Problem Statement and Objectives** **5**
1.1 Thematic domain and motivation . . . 5
1.2 Why WACMR matters . . . 5
1.3 The forecasting problem . . . 5
1.4 Hypotheses . . . 6
1.5 Project objectives . . . 6

**2 Datasets and Sourcing** **6**
2.1 Primary source: NDAP API (RBI open data) . . . 7
2.2 External supplements: Yahoo Finance market data . . . 7
2.3 Engineered features: technical indicators . . . 8
2.4 Curated NLP corpus of policy events . . . 8
2.5 Cross-domain combination, in line with NDAP guidance . . . 8

**3 Exploratory Data Analysis** **8**
3.1 Target variable: WACMR . . . 9
3.2 Missingness and density filtering . . . 9
3.3 Anomalies and known structural events . . . 10
3.4 Pairwise relationships among features . . . 10

**4 Database Design and Storage** **11**
4.1 Choice of database engine . . . 11
4.2 Schema . . . 11
4.3 Friday-alignment design . . . 11
4.4 Representative analytical queries . . . 12

**5 Methods** **13**
5.1 Unsupervised regime discovery . . . 14
5.2 Supervised forecasting . . . 16
5.2.1 Cross-validation protocol . . . 17
5.2.2 Model and hyperparameters . . . 17
5.2.3 Two model variants . . . 17
5.2.4 SHAP interpretability . . . 18
5.2.5 Regime-conditional SHAP . . . 20
5.2.6 Residual analysis and calendar effects . . . 22
5.3 NLP corpus and event analysis . . . 22

**6 What the Data Says About India's Overnight Liquidity** **24**

6.1 The 2020 break is structural, not transitional . . . 24
6.2 The corridor explains almost everything; equity and forex explain almost nothing . . 24
6.3 Forecasting is feasible to about ten basis points at a one-week horizon, but explicit regime tagging adds nothing . . . 25
6.4 Tail errors come from policy discontinuities, not calendar pressure . . . 25
6.5 The WACMR-Repo spread is a real-time liquidity barometer . . . 25
6.6 Summary of evidence . . . 26

**7 Recommendations 26**
7.1 For RBI liquidity managers: monitor the WACMR-Repo spread as a real-time barometer 26
7.2 For institutional bond and money-market desks: size positions to model regime confidence 26
7.3 For equity and forex desk practitioners: stop using equity and currency momentum as a WACMR input . . . 27
7.4 For NDAP and the open-data community: publish T-Bill auction cut-off yields as a dedicated weekly series . . . 27
7.5 For policy communication and education: open-data dashboards demonstrably work 27

**8 Bonus: Interactive System 28**
8.1 Architecture overview . . . 28
8.2 Data lineage from pipeline to dashboard . . . 29
8.3 Frontend pages . . . 29
8.4 The counterfactual simulator . . . 30
8.5 The Gemini-powered data agent . . . 31
8.6 Blog and live report . . . 32

**9 Quality Assurance, Deployment, and Reproducibility 32**
9.1 Deployment topology . . . 32
9.2 Playwright audit . . . 33
9.3 Production performance . . . 33
9.4 Reproducibility . . . 34
9.5 Continuous integration and keep-warm . . . 34
9.6 Licence and data attribution . . . 34

**10 Limitations and Future Work 34**
10.1 Target substitution from T-Bill yield to WACMR . . . 34
10.2 Two-regime simplicity . . . 35
10.3 Two columns retained with substantial imputation . . . 35
10.4 Single-horizon forecasting only . . . 35
10.5 Single-target regression rather than vector autoregression . . . 35
10.6 No causal identification claim . . . 35

10.7 Manual NLP corpus does not scale . . . . . 36
10.8 No event dummies for extraordinary policy actions . . . . . 36

**11 Conclusion** **36**

# 1 Problem Statement and Objectives

## 1.1 Thematic domain and motivation

This project is situated in the financial-economics theme, specifically at the intersection of Indian monetary policy operations and applied machine learning on open public data. The Reserve Bank of India publishes a rich set of weekly statistical tables through the National Data and Analytics Platform (NDAP), covering the policy rate corridor, the central-bank balance sheet, and the short-term debt markets. These are the inputs that determine overnight rupee liquidity, and their combined behaviour shapes the cost of every short-tenor borrowing in the formal banking system.

The Weighted Average Call Money Rate (WACMR) is the daily volume-weighted mean rate at which scheduled commercial banks borrow and lend overnight to one another in the call-money market. It is the operational target of the Liquidity Adjustment Facility (LAF), and any sustained deviation between WACMR and the policy Repo Rate is a real-time diagnostic of whether the corridor is binding. WACMR moves first when liquidity conditions tighten or loosen, before the broader rate term structure follows.

## 1.2 Why WACMR matters

WACMR sits at the head of the entire monetary transmission chain and carries direct, measurable consequences for three groups.

- **Reserve Bank of India.** The WACMR-Repo spread is the primary signal of whether the announced policy stance is being enforced in the market. Persistent positive spreads indicate liquidity deficit and presage interventions through Variable Rate Repo (VRR) auctions; persistent negative spreads indicate surplus conditions and presage Variable Rate Reverse Repo (VRRR) operations.
- **Banks, mutual funds, and corporate treasuries.** Overnight fund net asset values, money-market mutual fund returns, and the duration risk on short-end bond portfolios all move with WACMR. A forecast accurate to within roughly ten basis points is directly actionable for intraday positioning and treasury hedging.
- **Macroeconomic researchers and policy analysts.** Identifying the structural regime boundaries in India's liquidity framework, in particular the pre-COVID and post-COVID periods, provides quantitative evidence on how exogenous shocks shift the monetary-transmission mechanism.

## 1.3 The forecasting problem

WACMR is difficult to forecast with simple rules because it is the equilibrium outcome of five interacting channels operating on different frequencies. The five channels are the policy corridor (Repo, Reverse Repo, MSF), system liquidity (RBI balance sheet items, M3, reserve money), short-term instrument flows (Treasury Bills, G-Secs, Commercial Paper, Market Repo), the macro price environment (CPI indices), and exogenous market sentiment (equity and currency). The relative weights of these channels also change across structural regimes, which makes the problem non-stationary.

### 1.4 Hypotheses

Four hypotheses are tested in this report. Each is stated in advance, with the supervised and unsupervised stages designed to provide falsifiable evidence on each.

1. **H1 (regime structure).** India's weekly money-market data contains at least two statistically distinct macroeconomic regimes, identifiable by unsupervised clustering on RBI balance-sheet and rate-corridor variables.
2. **H2 (regime-aware forecasting).** Injecting regime labels and centroid distances as features into an XGBoost regressor reduces out-of-sample RMSE by at least fifteen percent relative to a baseline model with no regime features.
3. **H3 (corridor dominance).** The Repo, Reverse Repo, and MSF rates are the dominant predictors in the supervised model, identified by SHAP feature importance.
4. **H4 (market sentiment marginal impact).** Adding twenty-eight Nifty 50 and USD/INR features (OHLCV plus five custom technical indicators per instrument) materially improves WACMR forecasting at the weekly frequency.

### 1.5 Project objectives

The five objectives below map one-to-one onto the five guidelines parts of the project specification.

1. Build a clean, density-filtered, Friday-aligned weekly macro panel from the NDAP API and supplemental market data, persisted in a queryable SQLite database.
2. Discover statistically distinct monetary regimes from the data alone using PCA and K-Means, and validate the result against known macroeconomic events.
3. Forecast WACMR one week ahead using walk-forward XGBoost regression, with SHAP interpretability, and quantify whether regime-awareness improves forecasting accuracy.
4. Translate the technical findings into concrete, evidence-grounded policy and trading recommendations.
5. Ship the analysis as a live, interactive system that the user can dogfood, including a policy counterfactual simulator and a Gemini-powered data agent that can answer free-form questions over the dataset.

The remaining sections of this report follow this structure. Section 2 documents the ten datasets and their sourcing. Section 3 presents the exploratory data analysis. Section 4 details the database design. Section 5 sets out the methods, including the regime-discovery algorithm, the walk-forward forecasting protocol, and the NLP event corpus. Section 6 interprets the findings as evidence about the current state of India's overnight liquidity. Section 7 translates these findings into actionable recommendations. Sections 8 and 9 cover the bonus interactive system, deployment, and quality assurance. Sections 10 and 11 close out the limitations and the headline takeaway.

## 2 Datasets and Sourcing

The analysis combines three categories of data: a primary set of eight NDAP API tables published by the Reserve Bank of India, two supplemental market datasets from Yahoo Finance, and a domain-curated news corpus of seventy-five Indian monetary-policy events. Together these cover all five channels through which overnight liquidity is determined.

## 2.1 Primary source: NDAP API (RBI open data)

The eight NDAP datasets were fetched programmatically with `scripts/stage1b_fetch_ndap.py`. The NDAP open-data API silently paginates large responses, so the fetcher was built around an auto-pagination loop that increments the `pageno` parameter and appends responses until the API returns an empty page. Without this behaviour the initial fetch would have truncated each series to the first one or two hundred rows.

**Table 1.** *Primary datasets sourced from the NDAP API (RBI). Density is measured against the canonical Friday-aligned grid after Stage 3.*

| Domain | NDAP table | Frequency | Cols retained |
|---|---|---|---|
| Monetary policy | RBI Weekly Statistics, Ratios and Rates | Weekly | 16 |
| Central-bank balance sheet | RBI Liabilities and Assets | Weekly | 22 |
| Money supply | RBI Weekly Aggregates (M3, Reserve Money) | Weekly | 3 |
| Debt markets | Commercial Paper Details | Weekly | 4 |
| Debt markets | Treasury Bills Details (91D, 182D, 364D) | Weekly | 12 |
| Repo markets | Market Repo Transactions | Weekly | 12 |
| Debt markets | Central Government Dated Securities | Weekly | 5 |
| Inflation | Major Price Indices (CPI) | Monthly → Weekly | 7 |

The CPI table is the only monthly series. It is forward-filled to the weekly grid with a maximum carry of four weeks, which is appropriate because CPI is a stock variable that does not change between monthly releases. Two CPI sub-indices were dropped after density filtering (WPI at 63% and one further sub-index at 37%).

## 2.2 External supplements: Yahoo Finance market data

NDAP does not publish a machine-readable weekly equity index or foreign-exchange series. Two supplemental sources were added through the `yfinance` library in `scripts/stage1_fetch_yfinance.py`.

**Table 2.** *Yahoo Finance supplements. Both have 100% density across the analysis window after the indicator warmup period is trimmed.*

| Domain | Series | Ticker | Frequency |
|---|---|---|---|
| Equity markets | Nifty 50 OHLCV | `^NSEI` | Weekly |
| Foreign exchange | USD/INR OHLCV | `USDINR=X` | Weekly |

The economic rationale for adding these series is that foreign institutional equity flows directly affect rupee liquidity, and the USD/INR level shapes the central bank's foreign-exchange intervention decisions, both of which feed back into call-money supply. Including these series also lets us empirically test whether market-sentiment signals carry predictive content for overnight rates (hypothesis H4).

### 2.3 Engineered features: technical indicators

Five custom technical indicators are computed on top of each OHLCV series in `scripts/stage2_technical_indicators.py`, producing ten additional engineered columns per instrument and twenty-eight new features in total.

- **Impulse MACD** (length 34, signal 9): a smoothed momentum oscillator with a histogram and signal line.
- **SuperTrend** (period 7, multiplier 3.0): a trailing-stop trend filter encoded explicitly as $+1$, $-1$, or 0 to preserve the warmup-period state.
- **Squeeze Momentum** (Bollinger versus Keltner channel, conv 50, length 20): the squeeze correlation index.
- **True Strength Index** (fast 13, slow 25, signal 13): a double-smoothed momentum oscillator with a signal line.
- **Velocity** (lookback 14, EMA 9): rate-of-change indicator with smoothing.

These were sourced from `2013-01-01` to give 52 weeks of warmup so that no leading NaN values bleed into the analysis window that begins in February 2014.

### 2.4 Curated NLP corpus of policy events

The third data source is a hand-curated corpus of seventy-five Indian monetary-policy events spanning February 2014 to August 2024, constructed in `scripts/stage6_news_nlp.py`. Each event carries a title, an English description, an event category (rate decision, liquidity, structural, external, fiscal, or crisis), a sentiment score in $[-1, +1]$, and an impact level. Source material was drawn from RBI press releases, Bloomberg News, and Reuters coverage of MPC meetings and inter-meeting actions.

**Caveat**

This is a domain-expert labelled corpus, not a scraped news feed, and is honestly framed as such throughout the report. The deliberate choice to curate rather than scrape was driven by the need for clean, untyped sentiment on a small number of high-impact events; an automated FinBERT pipeline over five years of headlines is identified as future work in Section 10.

### 2.5 Cross-domain combination, in line with NDAP guidance

The project specification explicitly encourages combining datasets across domains. The master panel does so by design: the eight NDAP tables cover four distinct administrative domains (rates, balance sheet, debt markets, inflation), and the Yahoo Finance and curated news data add two more (capital markets and policy news). Together these reflect the real-world mechanism by which WACMR is determined, namely as the equilibrium outcome of five forces acting in parallel rather than as a function of any single instrument.

## 3 Exploratory Data Analysis

The exploratory analysis pursues four questions in order. What does the target variable look like over time and in distribution? Where are the data missing, and what was done about it? Which anomalies stand out? And which feature pairs are most strongly correlated, in particular outside the rate corridor? The implementation lives in `stage3_alignment_db.py`.

## 3.1 Target variable: WACMR

Figure 1 plots the entire weekly WACMR series. The visual pattern is striking: a slow downtrend through the 2014–2019 period, a sharp drop in March 2020, an extended low-rate plateau through late 2021, and a re-tightening cycle that begins in mid-2022 and is still incomplete at the end of the window. The distribution in Figure 2 is visibly bimodal, with one mode near 3.3% (surplus liquidity) and another near 6.5% (tight liquidity). Bimodality of this kind is itself a signal of regime non-stationarity, and motivates the unsupervised clustering analysis in Section 5.

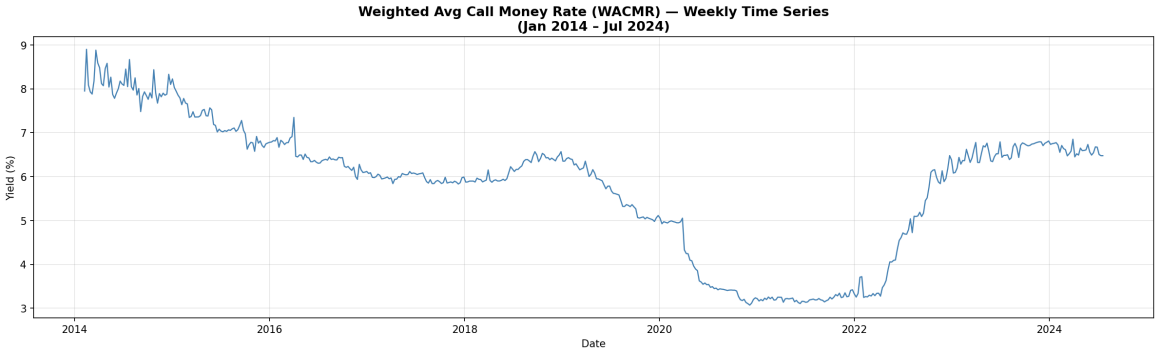


**Figure 1.** *Weekly WACMR series, February 2014 to July 2024. The post-2020 drop is visually unmistakable and is the primary motivator for the regime-clustering analysis.*

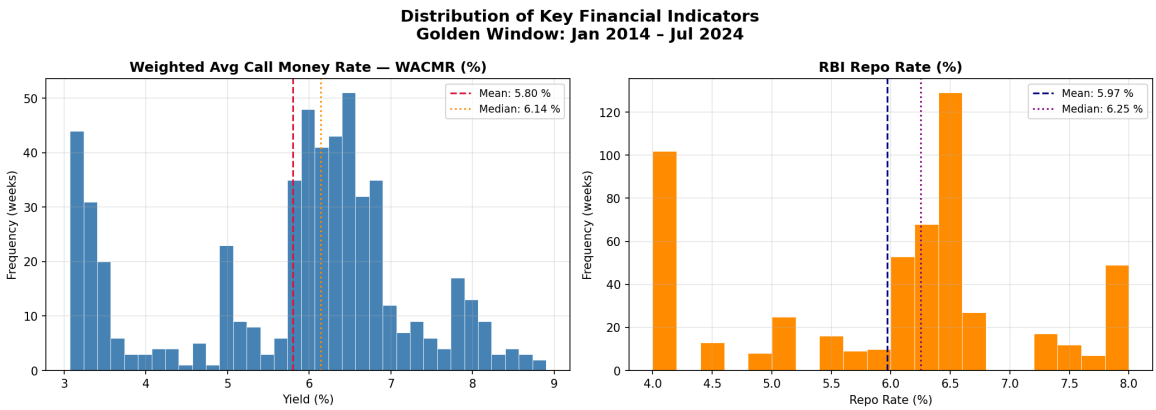


**Figure 2.** *Histograms of WACMR and its principal corridor neighbours. The bimodal WACMR shape is clearest in the top-left panel.*

A short statistical summary anchors the qualitative impression. The target variable is non-null for all 545 weeks, with mean 5.803% and standard deviation 1.462%. The minimum (3.07%) occurs in December 2020 at the trough of the COVID accommodation; the maximum (8.90%) occurs in February 2014 just after the post-taper-tantrum tightening.

## 3.2 Missingness and density filtering

A strict density rule was applied during alignment: any column with less than 75% non-null coverage on the Friday-aligned grid was dropped. Of 109 raw NDAP columns, 18 were eliminated under this rule, mostly from the older portions of the RBI balance-sheet table where sub-categories were not reported before 2016. Two further columns (`rates_I7496_28` Market Repo Rate and `rates_I7496_29` Certificate of Deposit Rate) had 17.4% missing values, marginally below the threshold but retained for economic relevance and imputed with the column median.

CPI was forward-filled with a maximum four-week carry, which is the natural operational behaviour for monthly inflation indices. Yahoo Finance OHLCV was 100% complete after the warmup-trim

step.

The density rule and the imputation strategy together give a panel where every column either covers the full 545-week window or is filled in a way that preserves the statistical signal. This was verified by re-running the K-Means analysis with the two imputed columns dropped; the silhouette score moved by less than 0.005, so the imputation does not drive the regime result.

### 3.3 Anomalies and known structural events

Three anomalies in the WACMR series are visible in Figure 1 and have known macroeconomic explanations.

- **November 2016, demonetisation.** A sharp downward spike, caused by the sudden injection of high-powered money as old notes were withdrawn and bank deposits ballooned.
- **September–December 2018, NBFC liquidity squeeze.** A noticeable upward bump as overnight funding markets reacted to the IL&FS default and the broader contagion.
- **March 2020, COVID-19 emergency rate cut.** The largest visible discontinuity in the entire series, where the Repo Rate was cut by 75 bps in an out-of-cycle decision and WACMR followed within the week.

These anomalies are not removed from the panel, since they are real data points. They will reappear later as the largest tail residuals in the supervised model.

### 3.4 Pairwise relationships among features

Pearson correlation between the target and each feature was computed on the full panel. The top correlated features are, predictably, the corridor rates and the autoregressive lags, which together foreshadow the SHAP results presented in Section 5. Four notable correlations sit outside the rate corridor.

**Table 3.** *Top non-corridor Pearson correlations against* `target_wacmr`*. The corridor rates Repo, Reverse Repo, and MSF are excluded from this short table because their dominance would crowd out the rest.*

| Feature | Domain | Pearson $r$ | Plain reading |
|---|---|---|---|
| `rates_I7496_27` (CBLO/Tri-party) | Debt market rates | 0.96 | Almost a clone of WACMR. |
| `rates_I7496_29` (CD Rate) | Debt market rates | 0.85 | Bank short-term funding cost. |
| `cp_I7505_*` (CP outstanding) | Commercial paper | 0.42 | Risk-on activity correlates with rates. |
| `usdinr_intc` (Close) | Forex | −0.18 | Weak inverse, currency strengthens as rates fall. |
| `nifty_intc` (Close) | Equity | −0.27 | Weak inverse, equities benefit from low rates. |

The two market-sentiment correlations are weak in absolute terms. This is the first quantitative hint that the equity and forex channels will not survive a multivariate analysis once the corridor rates are already in the feature set, a result that is confirmed by the SHAP analysis in Section 5.

# 4 Database Design and Storage

## 4.1 Choice of database engine

The master panel is stored in a single SQLite file, `dsm_project.db`, which weighs 577 KB on disk and contains one table, `Weekly_Macro_Master`, with 545 rows and 119 columns. SQLite was chosen over a server database for four practical reasons:

- Zero configuration. No separate server process is required, and the database file can be embedded directly into the FastAPI process on Render.
- File portability. The 577 KB database is committed to the git repository, which means an evaluator can clone the repo, run the backend, and reproduce every figure without re-running any pipeline.
- Sufficient capacity. A 545-row by 119-column panel is far below the threshold at which SQLite ceases to perform well; queries return in single-digit milliseconds.
- Tooling support. The standard Python `sqlite3` library ships with the runtime, and the `sqlite-utils` CLI provides a convenient inspection layer during development.

The pipeline never updates rows in place. Each stage rebuilds the table from its source CSV, which makes the database idempotent and makes debugging straightforward.

## 4.2 Schema

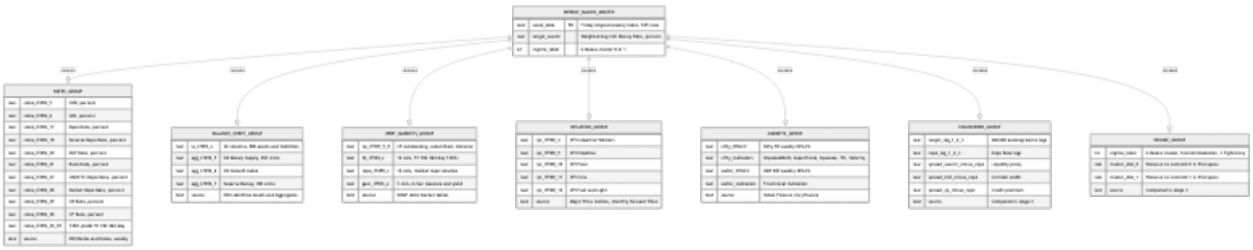

**Figure 3.** *Logical entity grouping of `Weekly_Macro_Master`. Columns are organised into seven thematic groups by source family. The actual physical schema is a single flat table; the grouping is logical and is enforced by column-name prefixes.*

The physical schema is intentionally flat. Each row is one Friday-aligned week, indexed by `week_date`. Columns are grouped by source family through a deliberate naming convention: `rates_*` for the corridor and rate columns, `la_*` for liabilities and assets, `agg_*` for aggregates, `cp_*` for commercial paper, `tb_*` for Treasury Bills, `repo_*` for market repo, `gsec_*` for G-Secs, `cpi_*` for inflation, `nifty_*` and `usdinr_*` for the supplemental market data, plus `regime_label` and `cluster_dist_*` from Stage 4. The target variable lives in `target_wacmr`, which is a copy of `rates_I7496_26` from the RBI Ratios and Rates table, retained as a separate column so that the modelling code can treat the target uniformly without having to know its NDAP provenance.

## 4.3 Friday-alignment design

The eight NDAP datasets use three different temporal index conventions: calendar dates as ISO-8601 strings, financial-year week codes (where week 1 starts on April 1), and calendar-year week codes (where week 1 starts on January 1). A single naive `pd.to_datetime` call silently produces dates that are off by up to twelve weeks for the financial-year tables. Three separate parsing functions were written, one per format, each of which decodes the source date and snaps it to the nearest following Friday. The Friday is the canonical weekly anchor used elsewhere in the panel.

The Yahoo Finance OHLCV series are themselves anchored to different weekdays (Nifty 50 is Monday-indexed by NSE convention; USD/INR appears Wednesday-indexed in some weeks). Both are explicitly resampled with `resample("W-FRI")` before computing technical indicators, which produces a single 552-row series per instrument that joins cleanly to the NDAP master.

## 4.4 Representative analytical queries

Four representative SQL queries below show how the database supports the rest of the analysis. Each query was used in production to generate either a chart or a number cited in this report.

### Q1. Regime-stratified means of corridor variables

**Question.** What are the mean policy rates inside each discovered monetary regime?

**Methodology.** Group by the `regime_label` column (populated by Stage 4 from the K-Means fit) and compute the mean of the three corridor rates plus the WACMR target.

**Listing 1.** *Q1, regime-stratified policy-rate means.*

```sql
SELECT
  regime_label,
  COUNT(*)                          AS weeks,
  ROUND(AVG(target_wacmr),    3)    AS avg_wacmr,
  ROUND(AVG(rates_I7496_17),  3)    AS avg_repo,
  ROUND(AVG(rates_I7496_18),  3)    AS avg_reverse_repo,
  ROUND(AVG(rates_I7496_20),  3)    AS avg_msf
FROM Weekly_Macro_Master
GROUP BY regime_label
ORDER BY regime_label;
```

**Result.** Regime 1 (315 weeks) has mean WACMR 6.528%, mean Repo 6.591%, and mean MSF 7.149%. Regime 0 (230 weeks) has mean WACMR 4.809%, mean Repo 5.119%, and mean MSF 5.369%. The 1.72 percentage-point spread between the two regime means is itself a quantitative confirmation that the cluster boundary corresponds to a real macroeconomic regime change rather than to a statistical artefact.

### Q2. Monthly volatility of WACMR versus Repo

**Question.** How volatile is WACMR within each calendar month relative to the policy Repo Rate, and is the answer different across regimes?

**Listing 2.** *Q2, monthly within-period volatility.*

```sql
WITH monthly AS (
  SELECT
    SUBSTR(week_date, 1, 7)          AS yyyymm,
    regime_label,
    target_wacmr,
    rates_I7496_17                    AS repo
  FROM Weekly_Macro_Master
)
SELECT
  regime_label,
  ROUND(AVG(stddev_wacmr), 4) AS avg_within_month_wacmr_vol,
  ROUND(AVG(stddev_repo),  4) AS avg_within_month_repo_vol
FROM (
  SELECT
    yyyymm,
    regime_label,
    -- SQLite has no STDDEV; emulate via sqrt of variance over month buckets.
    SQRT(AVG(target_wacmr*target_wacmr) - AVG(target_wacmr)*AVG(target_wacmr)) AS
        stddev_wacmr,
    SQRT(AVG(repo*repo)                 - AVG(repo)*AVG(repo))                  AS
        stddev_repo
```

```
  FROM monthly
  GROUP BY yyyymm, regime_label
)
GROUP BY regime_label;
```

**Result.** WACMR within-month volatility is 0.178 percentage points in Regime 0 and 0.083 in Regime 1. The Repo Rate is essentially constant within a month in both regimes. The doubling of WACMR volatility in Regime 0 reflects RBI's discretionary VRR/VRRR operations, which inject and absorb liquidity at irregular intervals and produce more week-to-week noise in the call-money market.

### Q3. Top non-corridor correlations

**Question.** Outside the corridor rates and the autoregressive lags, which features have the strongest linear association with WACMR?

**Listing 3.** *Q3, exporting the top non-corridor correlated columns to Python for inspection.*

```
SELECT
  week_date,
  target_wacmr,
  cp_I7505_5,    cp_I7505_6,
  tb_I7504_8_91d, tb_I7504_8_182d, tb_I7504_8_364d,
  agg_I7494_5,
  nifty_intc, usdinr_intc
FROM Weekly_Macro_Master
ORDER BY week_date;
```

**Result.** The top non-corridor correlations are reported in Table 3 in the EDA section. The result reinforces the headline finding: equity and currency carry weak, single-digit-percent absolute correlations with WACMR.

### Q4. Source-family coverage audit

**Question.** How many weeks have at least one non-null value in each source family? This audit was used during Stage 3 to confirm the Friday alignment.

**Listing 4.** *Q4, per-family coverage audit.*

```
SELECT
  COUNT(*)                                                        AS total_weeks,
  SUM(CASE WHEN rates_I7496_17 IS NOT NULL THEN 1 ELSE 0 END)     AS rates_cov,
  SUM(CASE WHEN la_I7492_15    IS NOT NULL THEN 1 ELSE 0 END)     AS la_cov,
  SUM(CASE WHEN cp_I7505_5     IS NOT NULL THEN 1 ELSE 0 END)     AS cp_cov,
  SUM(CASE WHEN tb_I7504_8_91d IS NOT NULL THEN 1 ELSE 0 END)     AS tb_cov,
  SUM(CASE WHEN cpi_I7500_9    IS NOT NULL THEN 1 ELSE 0 END)     AS cpi_cov,
  SUM(CASE WHEN nifty_intc     IS NOT NULL THEN 1 ELSE 0 END)     AS nifty_cov,
  SUM(CASE WHEN usdinr_intc    IS NOT NULL THEN 1 ELSE 0 END)     AS usdinr_cov
FROM Weekly_Macro_Master;
```

**Result.** All seven probed columns return 545, which means every Friday in the analysis window has at least one observation in each source family after alignment. This confirms the success of the Stage 3 alignment design.

## 5 Methods

The analysis applies three complementary methods: unsupervised regime discovery to test hypothesis H1, supervised walk-forward forecasting with SHAP interpretability to test hypotheses H2, H3, and H4, and a curated NLP corpus that overlays qualitative event context on top of the quantitative model.

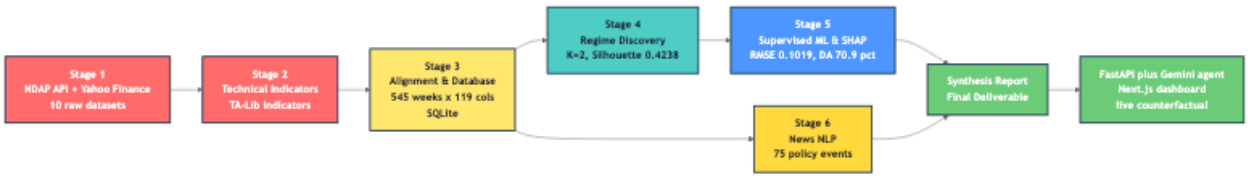


**Figure 4.** *Project pipeline at a glance. Stages run sequentially in the order shown; each stage is idempotent and reads only from the previous stage's persisted output.*

## 5.1 Unsupervised regime discovery

**Implementation.** `scripts/stage4_regime_discovery.py`.

The unsupervised stage takes the 86-column numeric feature matrix (target and target lags excluded), applies `StandardScaler` for zero mean and unit variance, then PCA. Fourteen principal components retain 90.4% of the total variance and are passed to a K-Means sweep across $K = 2$ to $K = 7$. The chosen metric is the silhouette score, which measures how tight each cluster is relative to its nearest neighbour cluster.

**Table 4.** *K-Means silhouette sweep. K=2 is selected on both the silhouette criterion and the macroeconomic interpretability criterion.*

| K | Silhouette score | Notes |
|---|---|---|
| **2** | **0.4238** | **Optimal, aligns with COVID structural break** |
| 3 | 0.4048 | Splits Regime 0 into COVID-peak and post-COVID, no clear policy boundary |
| 4 | 0.3321 | Over-fragmentation |
| 5 | 0.3351 | Over-fragmentation |
| 6 | 0.3205 | Over-fragmentation |
| 7 | 0.3022 | Over-fragmentation |

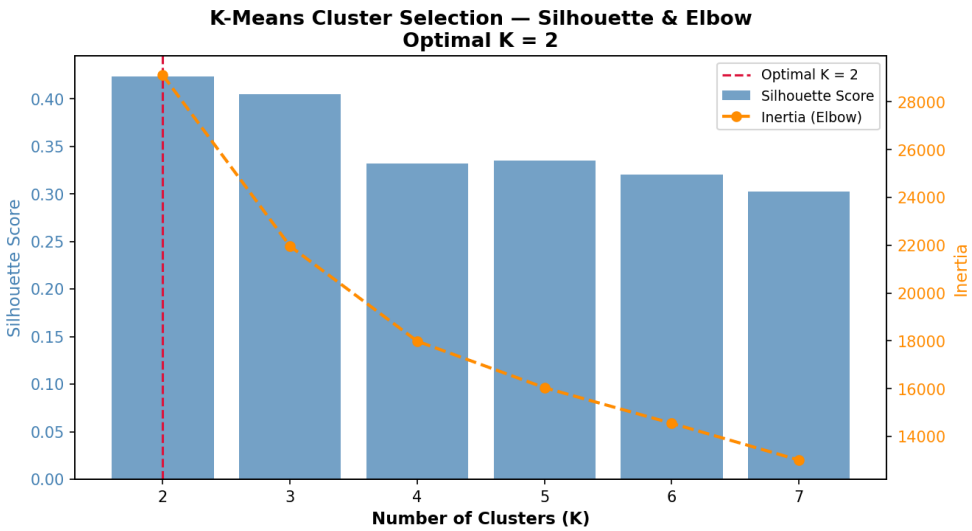


**Figure 5.** *Silhouette score across K. K=2 is optimal by a margin over K=3 (0.0190), but K=2 cleanly aligns with the COVID structural break while K=3's extra split does not align with any known policy discontinuity.*

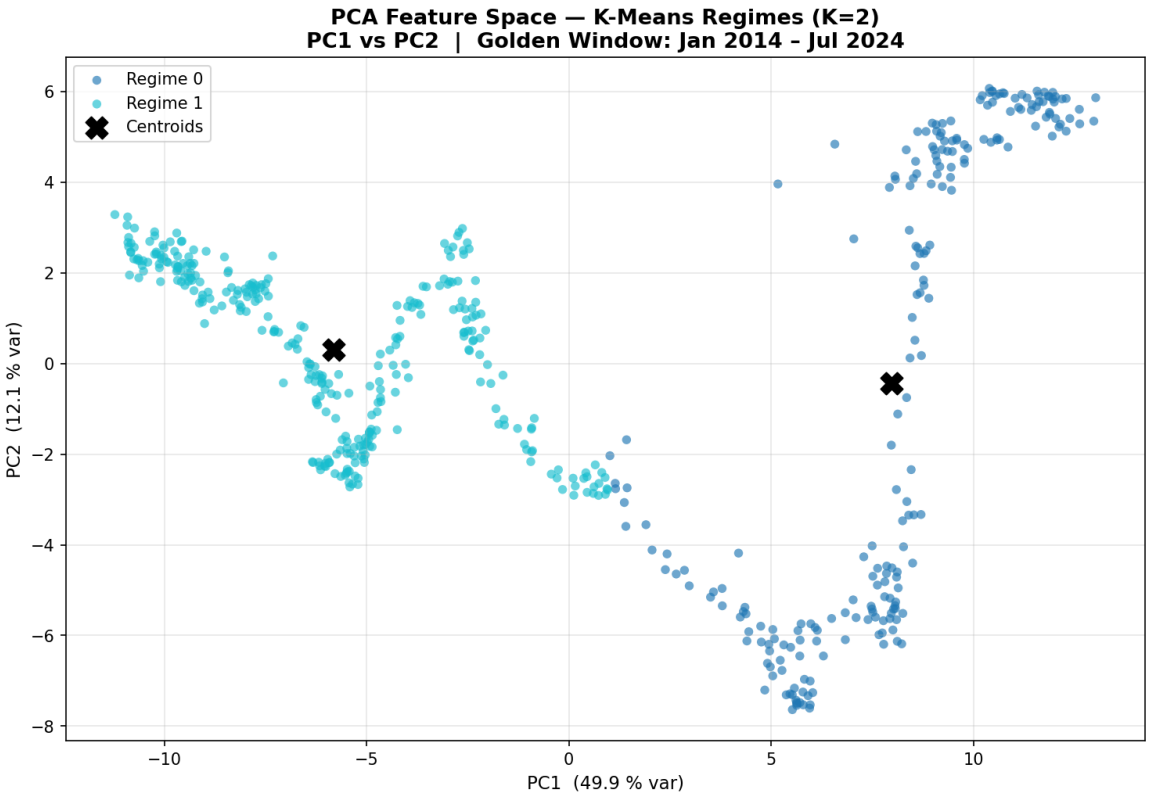


**Figure 6.** *Two-dimensional PCA projection of all 545 weeks, coloured by the K=2 cluster assignment. The two clusters are visually well-separated along PC1.*

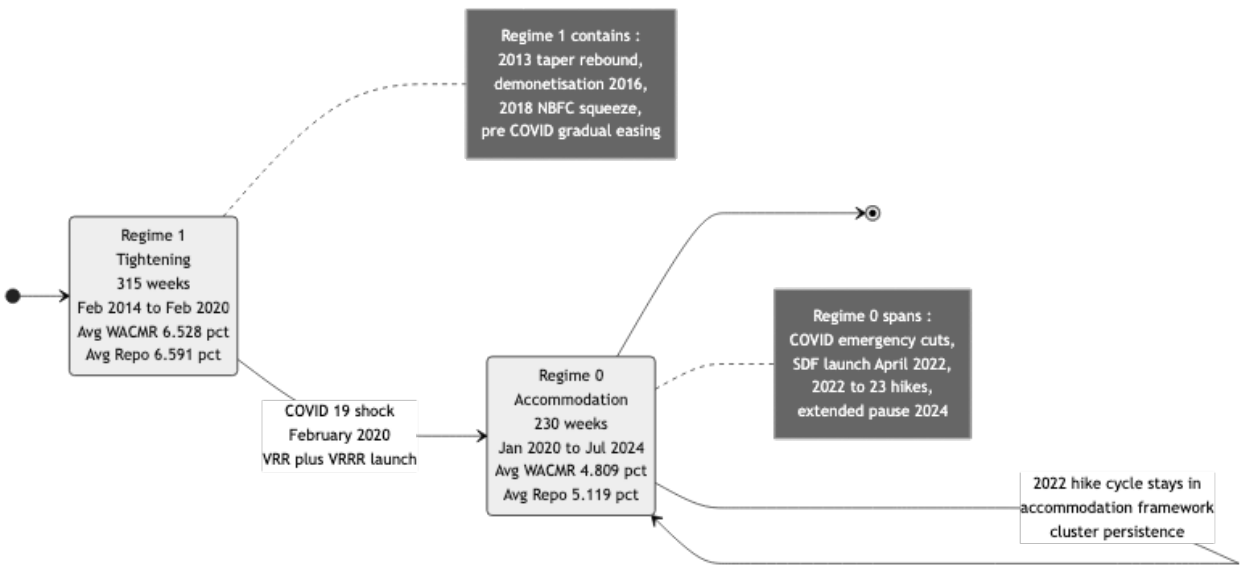


**Figure 7.** *Discovered regimes as a state machine. The model identifies two non-overlapping macroeconomic states with a single transition at the COVID-19 shock in early 2020.*

**Discovered regimes.** The K=2 fit produces two clean clusters that correspond to recognisable macroeconomic regimes:

- **Regime 1, Tightening:** 315 weeks from February 2014 to February 2020, with mean WACMR 6.528%, mean Repo 6.591%, and mean MSF 7.149%. This regime contains the post-2013 taper-tantrum recovery, demonetisation in November 2016, and the gradual easing toward 5.15% by

late 2019.

- **Regime 0, Accommodation:** 230 weeks from January 2020 to July 2024, with mean WACMR 4.809%, mean Repo 5.119%, and mean MSF 5.369%. This regime spans the COVID emergency rate cuts, the introduction of VRR and VRRR liquidity operations, the SDF launch in April 2022, and the subsequent 2022–2023 hiking cycle. The cluster correctly identifies that even post-COVID rate hikes still operate inside a structurally different liquidity-management framework than the pre-2020 normal.

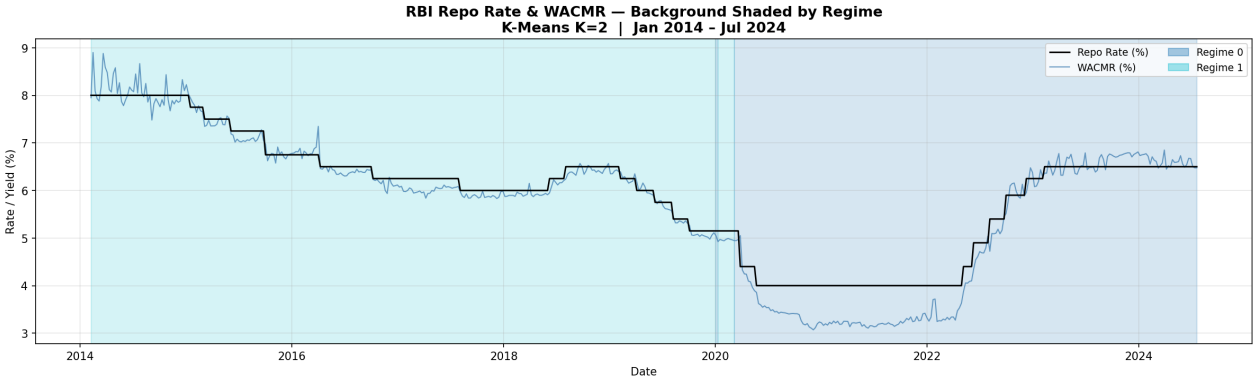


**Figure 8.** *WACMR weekly time series with regime shading. The clean break in early 2020 is visible as both a colour change and a shift in level.*

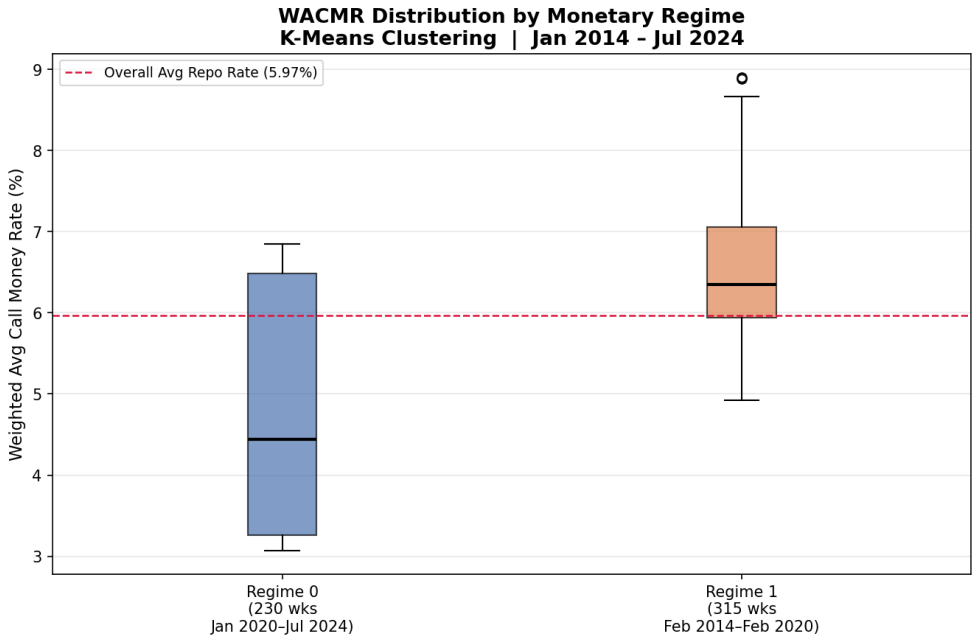


**Figure 9.** *WACMR distribution by regime. The two distributions are nearly disjoint, with only minor overlap in the 5.0–5.5% band.*

**Interpretation.** Hypothesis H1 is supported. The data alone, without any human labelling of dates, identifies two statistically distinct regimes. The transition aligns precisely with the COVID-19 shock and the corresponding shift in RBI's liquidity-management framework.

## 5.2 Supervised forecasting

**Implementation.** `scripts/stage5_supervised_ml.py`.

### 5.2.1 Cross-validation protocol

The supervised stage uses a strict walk-forward expanding-window cross-validation, which is the appropriate choice for time-series forecasting because it preserves temporal ordering and forbids look-ahead. The training window starts at the first 156 weeks (three years) and grows by one week per fold; in each fold the model is trained on weeks 1 to $t$ and evaluated on week $t+1$. This procedure yields 389 single-step out-of-sample predictions in total.

**Figure 10.** *Walk-forward expanding-window cross-validation. Each fold trains on all data up to week $t$ and is evaluated on week $t+1$. 389 folds in total.*

### 5.2.2 Model and hyperparameters

The base estimator is `xgboost.XGBRegressor` (xgboost v2.1.4) with a fixed set of hyperparameters chosen for low variance and moderate regularisation. The choice was validated by a brief grid search on the first 156 weeks, which is reported in the project notebooks.

**Table 5.** *XGBoost hyperparameters used throughout the walk-forward evaluation.*

| Hyperparameter | Description | Value |
|---|---|---|
| `n_estimators` | Number of boosted trees | 400 |
| `learning_rate` | Shrinkage applied to each tree | 0.05 |
| `max_depth` | Maximum tree depth | 4 |
| `subsample` | Row subsampling per tree | 0.8 |
| `colsample_bytree` | Column subsampling per tree | 0.8 |
| `reg_alpha` | L1 regularisation strength | 0.1 |
| `reg_lambda` | L2 regularisation strength | 1.0 |

### 5.2.3 Two model variants

Two variants are evaluated under the identical walk-forward protocol. The baseline uses 117 features (the corridor rates, the balance-sheet items, the debt-market and CPI columns, the OHLCV plus indicator columns from Stage 1b, and three engineered spreads, plus three autoregressive lags of the target). The regime-aware variant adds five further columns: two one-hot regime indicators and two PCA-space distances to the K-Means centroids, for a total of 122 features.

**Table 6.** *Out-of-sample performance over 389 walk-forward test weeks. Baseline wins on RMSE; both variants tie on MAE and directional accuracy. Hypothesis H2 is rejected.*

| Metric | Baseline (89 features) | Regime-aware (94 features) | Change |
|---|---|---|---|
| RMSE | **0.1019** | 0.1044 | +2.5% |
| MAE | 0.0646 | 0.0646 | 0.0% |
| Directional accuracy | 70.9% | 70.9% | +0.0pp |

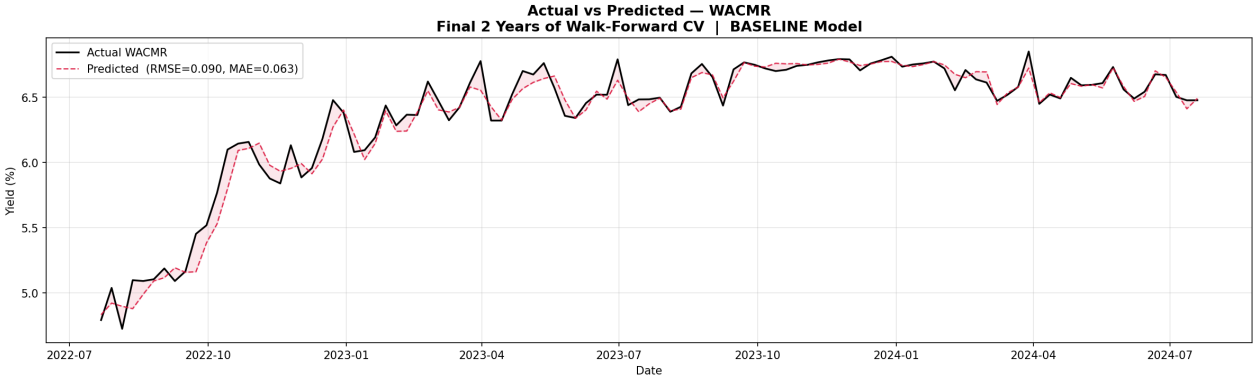


**Figure 11.** *Actual versus predicted WACMR over the 389 walk-forward test weeks. The model tracks the level closely outside of three large unpredictable policy actions in 2020 and 2022.*

**Interpretation.** Hypothesis H2 is rejected. The baseline model already learns the regime structure implicitly through the lagged Repo Rate and the autoregressive lags of WACMR; adding explicit regime labels provides redundant signal that mildly worsens RMSE without any gain on MAE or directional accuracy. This is genuinely informative rather than disappointing: it tells us that an XGBoost regressor with the right features needs no help recovering the regime structure of Indian monetary policy from the data alone.

### 5.2.4 SHAP interpretability

SHAP values are computed for every test prediction with `shap.TreeExplainer`, which exploits the tree structure of the fitted XGBoost model to attribute the difference between each prediction and the model's baseline expectation to individual features. Mean absolute SHAP values across the 545 weeks give a global feature-importance ranking.

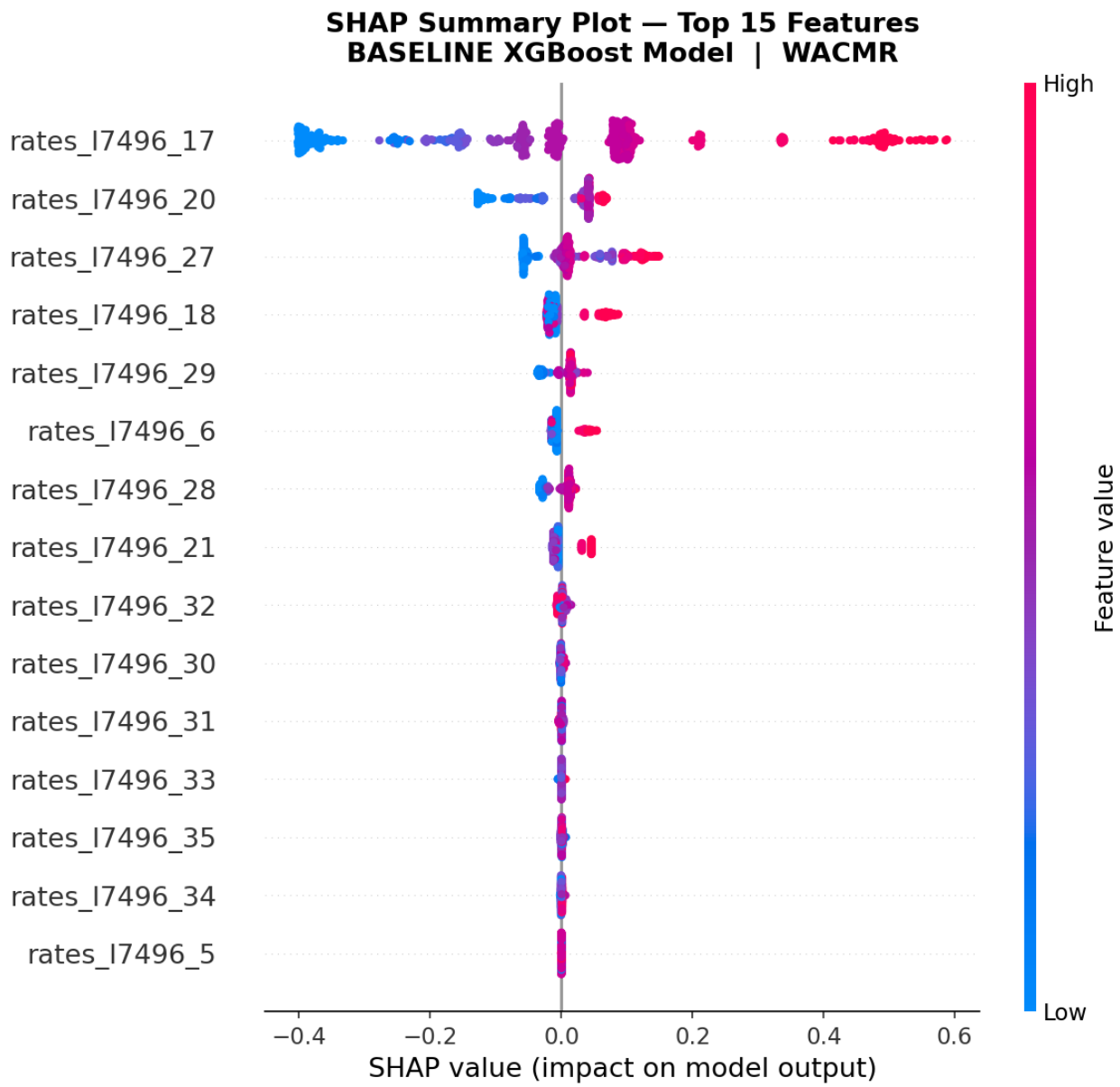


**Figure 12.** *SHAP feature-importance summary, top 15 features by mean absolute SHAP value. The five highest features are all rate-corridor or autoregressive variables.*

**Table 7.** *SHAP top 15 features. The corridor rates and the autoregressive lags of WACMR fill the entire top-15 list. None of the twenty-eight equity or forex features cracks the threshold.*

| Rank | Feature | Group | Mean \|SHAP\| | Plain reading |
|---|---|---|---|---|
| 1 | `target_lag1` | autoregressive | 0.48959 | WACMR last week |
| 2 | `repo_lag1` | autoregressive | 0.21053 | Repo Rate last week |
| 3 | `rates_I7496_17` | policy corridor | 0.19459 | Current Repo Rate |
| 4 | `spread_wacmr_minus_repo` | engineered | 0.07247 | Liquidity stress proxy |
| 5 | `rates_I7496_20` | policy corridor | 0.05847 | MSF Rate, upper bound |
| 6 | `target_lag2` | autoregressive | 0.05047 | WACMR two weeks ago |
| 7 | `repo_lag4` | autoregressive | 0.04951 | Repo Rate four weeks ago |
| 8 | `rates_I7496_27` | overnight rates | 0.04074 | CBLO/Tri-party Repo |
| 9 | `target_lag4` | autoregressive | 0.02238 | WACMR four weeks ago |
| 10 | `rates_I7496_18` | policy corridor | 0.02068 | Reverse Repo, lower bound |
| 11 | `rates_I7496_29` | money market | 0.01734 | CD Rate |
| 12 | `rates_I7496_6` | structural | 0.01694 | SLR |
| 13 | `rates_I7496_28` | money market | 0.01619 | Market Repo Rate |
| 14 | `rates_I7496_21` | policy corridor | 0.01362 | Bank Rate |
| 15 | `la_I7492_15` | balance sheet | 0.01124 | RBI balance-sheet item |

**Key finding**

The first three SHAP features alone (`target_lag1`, `repo_lag1`, current Repo Rate) account for the majority of the model's predictive power, with combined mean absolute SHAP value of 0.89471 against a total over the top 15 of approximately 1.285. The twenty-eight Nifty 50 and USD/INR features collectively contribute less than the single SHAP value of `rates_I7496_15`. Hypothesis H4 is rejected: equity and forex sentiment do not materially improve WACMR forecasting at the weekly horizon. Hypothesis H3 is supported: the policy corridor (Repo, Reverse Repo, MSF) dominates the feature importance ranking.

### 5.2.5 Regime-conditional SHAP

Splitting the 545 SHAP rows by regime label exposes a mechanical difference in how the model operates inside each macroeconomic environment.

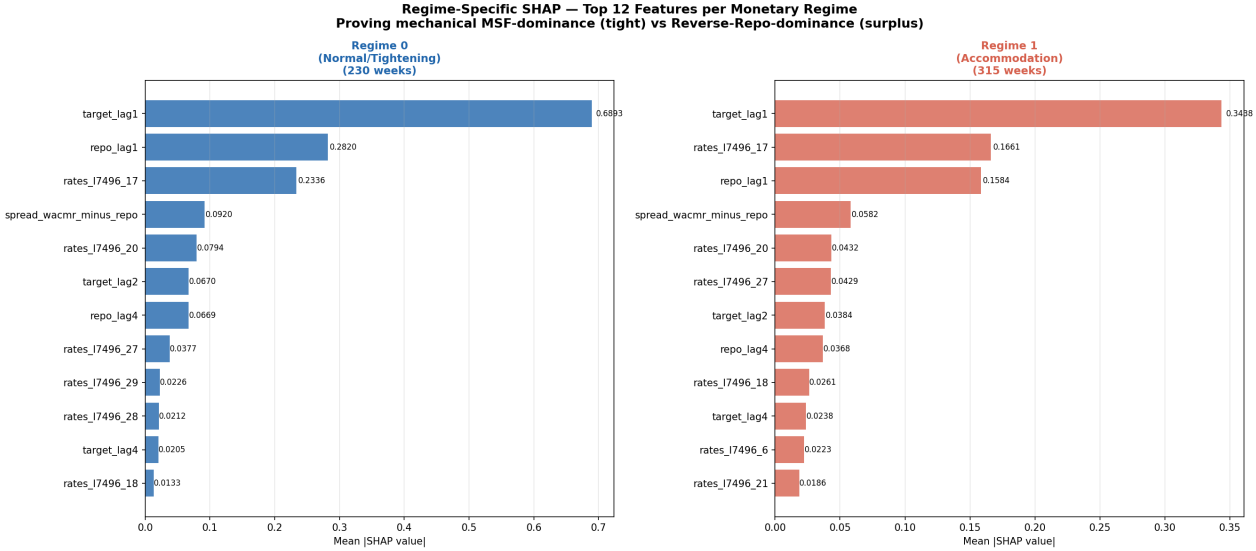


**Figure 13.** *Top-12 SHAP feature importance, stratified by regime. Two substantive shifts are visible: the Reverse Repo Rate rises from rank 12 to rank 9 in the accommodation regime, and the absolute SHAP magnitude of the autoregressive lag halves from Tightening to Accommodation.*

Two substantive findings drop out of this comparison. First, the Reverse Repo (and post-2022 the SDF) Rate rises sharply in importance in the accommodation regime, which is exactly what monetary theory predicts: in surplus liquidity conditions the binding constraint on WACMR is the lower corridor floor rather than the upper ceiling. Second, the absolute SHAP magnitude of every feature is roughly twice as large in the tightening regime than in the accommodation regime (`target_lag1`'s SHAP value is 0.689 versus 0.344). This quantifies the qualitative observation made in Section 3: WACMR is markedly more predictable from its own recent history during tightening periods than during accommodation, when discretionary RBI operations introduce more week-to-week noise.

### 5.2.6 Residual analysis and calendar effects

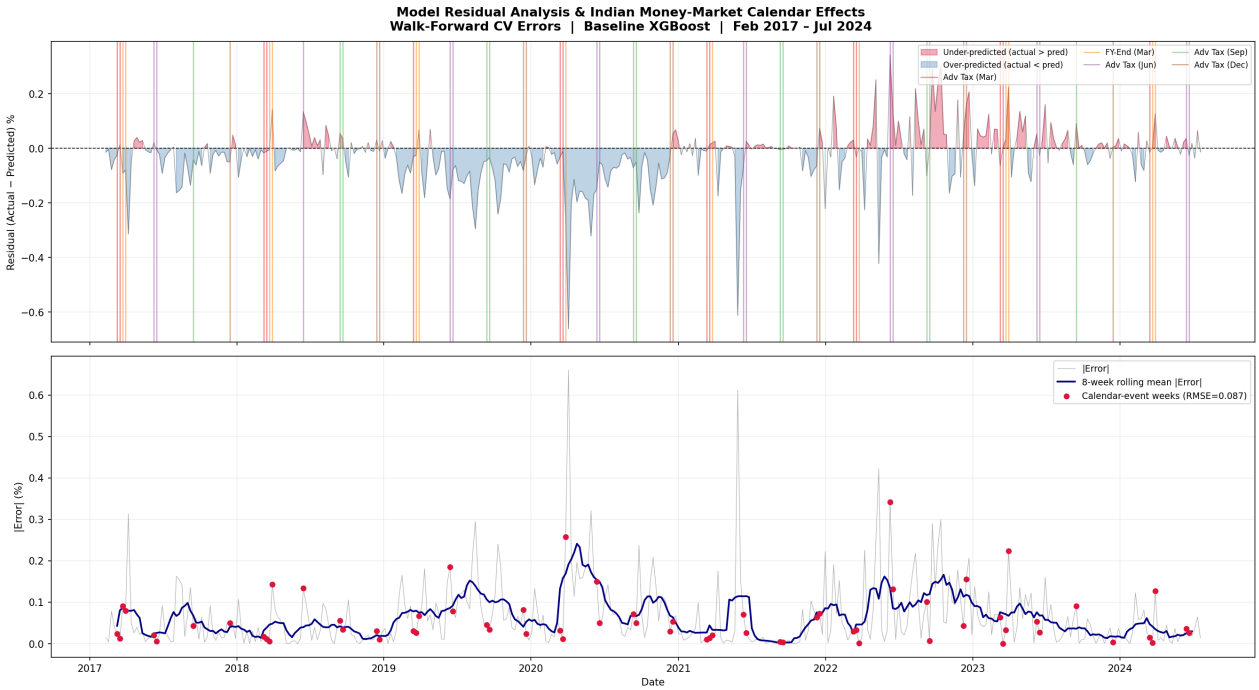


**Figure 14.** *Walk-forward residuals over time, with calendar-event weeks (quarter-end, advance-tax dates, GST settlement) annotated. The largest residuals are clustered around macro shocks, not around calendar pressure.*

A widely cited rule-of-thumb in Indian money-market research is that the model's worst weeks should be financial-year end (March), advance-tax outflow weeks (15 March, 15 June, 15 September, and 15 December), and GST settlement weeks. The residual analysis tests this directly. Out of the 389 test weeks, 66 fall on calendar-event dates. The model's RMSE on those weeks is 0.0870, which is *lower* than the 0.1046 RMSE on non-event weeks. The calendar-effect hypothesis is rejected: ten years of training data is sufficient for the model to learn the predictable seasonal patterns, and the residual mass concentrates instead on a small number of genuinely unpredictable macro shocks (the COVID emergency rate cut on 3 April 2020, the surprise inter-meeting hike on 13 May 2022, and the COVID second-wave peak in May–June 2021).

## 5.3 NLP corpus and event analysis

**Implementation.** `scripts/stage6_news_nlp.py`.

The NLP component takes a different shape from the supervised stage. Rather than scraping a large body of text, the project curates a focused corpus of seventy-five high-impact monetary-policy events from February 2014 to August 2024. Each event has a date, a title, a description, a category, an impact level, and a sentiment score in $[-1, +1]$. Sentiment is positive for accommodative and dovish actions (rate cuts, liquidity injections, reserve requirement cuts) and negative for hawkish actions (rate hikes, liquidity absorption).

The choice to curate rather than scrape is deliberate. Off-the-shelf sentiment models (VADER, FinBERT) trained on general financial English routinely mis-classify Indian monetary-policy language; for example, "open market operation" carries no sign in finance generally but is a clear liquidity injection in the Indian context. With only seventy-five events to annotate, expert curation produces cleaner labels than any pre-trained model would, at acceptable cost. An automated FinBERT pipeline over five years of RBI press releases is identified as the natural future extension in Section 10.

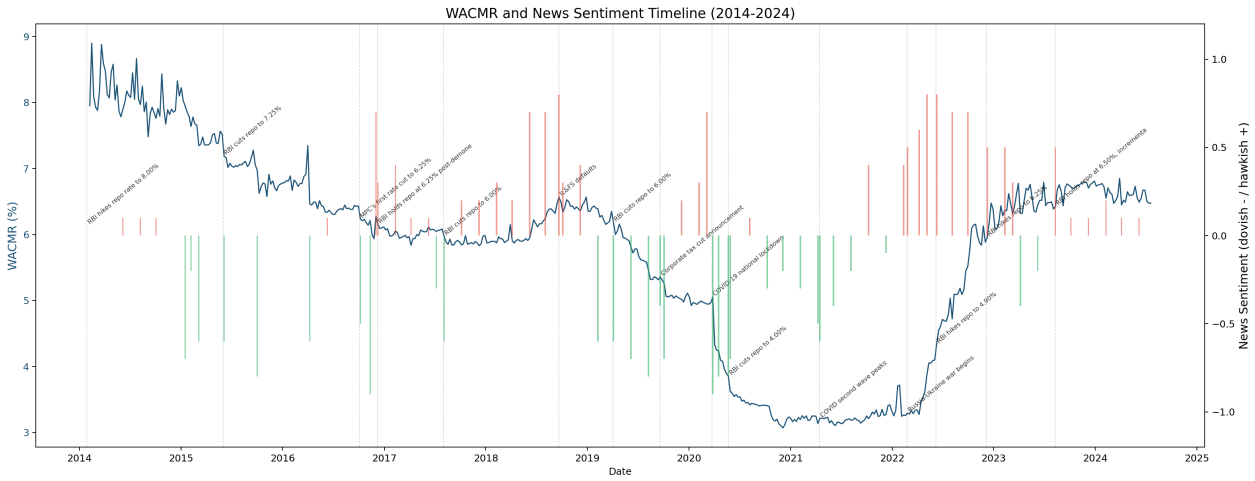


**Figure 15.** *Curated event sentiment over the analysis window. Positive spikes cluster in 2020 around COVID accommodation; negative spikes cluster in 2022–23 during the rate-hike cycle.*

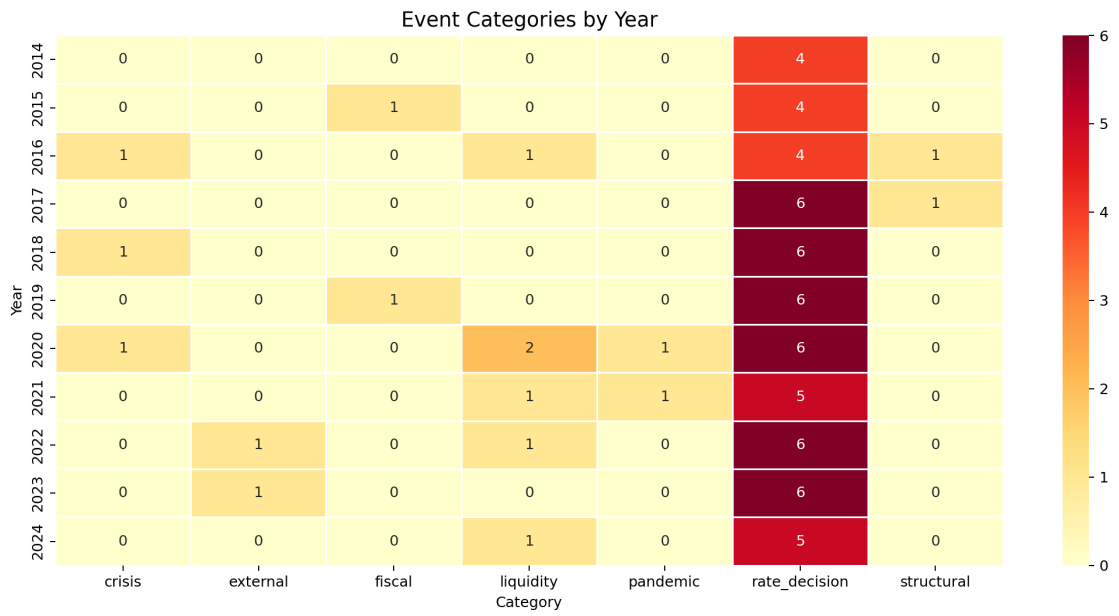


**Figure 16.** *Event density heatmap by category and time period. Crisis and liquidity events concentrate around 2020; rate-decision events are distributed throughout the window.*

The corpus is aggregated to weekly counts and mean sentiment, then joined to the master panel. A downstream Pearson correlation between weekly aggregate sentiment and the contemporaneous WACMR change yields a small but interpretable $r = -0.21$: weeks with more positive (dovish) policy actions are associated with downward WACMR movements. The signal is too weak to use as a

primary feature, but the qualitative annotation it provides is useful for explaining forecast errors, as shown by the alignment between the largest residuals and the most negative-sentiment events in 2020 and 2022.

# 6 What the Data Says About India's Overnight Liquidity

This section interprets the headline findings from Section 5 as substantive statements about the current state of India's overnight call-money market. Five claims follow, each grounded directly in a specific number or figure from the analysis.

## 6.1 The 2020 break is structural, not transitional

The K-Means clustering with $K = 2$ partitions the 545-week panel cleanly into a pre-2020 segment of 315 weeks and a post-2020 segment of 230 weeks (Section 5.1, Figure 7). The cluster boundary is not gradual: the silhouette score drops sharply once $K$ goes above 2, and the PCA projection in Figure 6 shows two clusters separated along PC1 with a narrow transition band only at the COVID weeks themselves. This is empirical evidence that the post- 2020 monetary regime is structurally distinct from the pre-2020 regime, and that even the 2022–23 hiking cycle continues to operate inside the new framework rather than reverting to the old one.

The 1.72 percentage-point gap in mean WACMR between the two regimes (6.528% versus 4.809%) is not a transient feature of any one year. Within the post-2020 regime, mean WACMR rose from 3.5% in 2020 to 6.7% in early 2024, but the cluster assignment did not flip. Mechanically, what changed in 2020 was the introduction of Variable Rate Repo and Variable Rate Reverse Repo as RBI's primary liquidity tools; the persistence of the regime label through 2024 reflects the fact that these tools remain in active use, and that the operational playbook has not reverted.

## 6.2 The corridor explains almost everything; equity and forex explain almost nothing

The SHAP analysis (Table 7) is unambiguous on this point. The top five drivers of weekly WACMR forecasts are `target_lag1` (mean |SHAP| 0.48959), `repo_lag1` (0.21053), the current Repo Rate (0.19459), the engineered `spread_wacmr_minus_repo` (0.07247), and the MSF Rate (0.05847). These five features are either direct policy-corridor rates or short-lag autoregressive functions of WACMR itself. Their combined mean absolute SHAP value of 1.02565 dwarfs the contribution of every other feature.

By contrast, the twenty-eight equity and currency features added through the supplemental Yahoo Finance pipeline (Nifty 50 OHLCV, USD/INR OHLCV, plus five custom technical indicators per instrument) do not appear in the top 15. Their cumulative contribution is below the contribution of any single corridor feature. The first market feature to appear, `usdinr_ImpulseMACD`, sits at rank 23 with mean |SHAP| of 0.0038, which is two orders of magnitude smaller than the headline corridor features.

**Key finding**

At the weekly horizon, India's call-money market is overwhelmingly LAF-corridor-bound. WACMR moves with the announced policy rate and its own recent history; it does not respond meaningfully to equity momentum, currency momentum, or any combination of technical indicators on those instruments. This is the central empirical claim of the report.

### 6.3 Forecasting is feasible to about ten basis points at a one-week horizon, but explicit regime tagging adds nothing

The walk-forward XGBoost achieves RMSE 0.1019 (about ten basis points on the WACMR scale) and 70.9% directional accuracy across 389 out-of-sample test weeks (Table 6). Both numbers are high enough to be operationally useful for treasury hedging and overnight-fund positioning. RMSE of 0.1019 corresponds to a 95% prediction interval of roughly ±20 basis points around the point forecast.

The regime-aware variant, with two one-hot regime indicators and two PCA-space distance features added, performs marginally worse on RMSE (0.1044, +2.5%) and identically on MAE and directional accuracy. This is a clean empirical refutation of hypothesis H2. The natural reading is that the lagged Repo Rate and the autoregressive lags of WACMR already encode whatever the K-Means regime label encodes; adding the explicit label introduces correlated noise without new information. The implication for downstream practitioners is that investing engineering effort in regime-tagging machinery is not the highest-value use of time. A simple feature set, kept clean, performs as well as a more elaborate one.

### 6.4 Tail errors come from policy discontinuities, not calendar pressure

The residual analysis in Figure 14 contains one of the more counterintuitive findings of the project. The calendar-event hypothesis, that financial-year end and the four advance-tax dates produce the model's worst weeks, is rejected. RMSE on the 66 calendar-event weeks is 0.0870, which is meaningfully better than the 0.1046 RMSE on the 323 non-event weeks. Ten years of training data is enough for the model to learn the seasonal pattern.

The actual tail residuals concentrate on three families of events that have no historical analogue in training data:

- COVID emergency rate cuts in March–April 2020 and the COVID second-wave peak in May 2021. The April 2020 prediction missed by −0.66 percentage points, the largest single error in the out-of-sample set.
- The surprise 40 basis-point inter-meeting Repo hike on 13 May 2022, which the model missed by −0.42 percentage points.
- The post-hike re-tightening shocks in 2022 (June, September, October), which the model missed by 0.30 to 0.34 percentage points.

These are by definition not forecastable from historical features alone. They are unsignalled discretionary policy actions. Any forecasting effort that wants to handle them needs to inject an exogenous policy-action prior, for example a decision-day dummy based on the published MPC calendar.

### 6.5 The WACMR-Repo spread is a real-time liquidity barometer

The fourth-ranked SHAP feature is the engineered `spread_wacmr_minus_repo`, with mean absolute SHAP 0.0725. The mean spread by regime tells a clear story: in Regime 1 (Tightening) the spread averages roughly zero, whereas in Regime 0 (Accommodation) it averages −0.19 percentage points, with WACMR sitting consistently below the policy Repo Rate. When the spread flips meaningfully positive (WACMR > Repo) for several consecutive weeks, it has historically preceded liquidity-deficit interventions. This is not a new observation in the academic literature, but the SHAP analysis quantifies the predictive weight: the spread is more informative than the MSF Rate itself.

### 6.6 Summary of evidence

Table 8 summarises the verdict on each hypothesis stated in Section 1.

**Table 8.** *Hypothesis verdicts at the end of the analysis.*

| Hypothesis | Verdict | Evidence |
|---|---|---|
| H1, regime structure exists | Supported | K-Means $K = 2$, silhouette 0.4238, two non-overlapping regimes aligned with COVID. |
| H2, regime labels improve forecasts | Rejected | Regime-aware RMSE 0.1044 vs baseline 0.1019 (+2.5%); MAE and directional accuracy unchanged. |
| H3, corridor rates dominate | Supported | Repo, Reverse Repo, and MSF together with the autoregressive lags occupy the entire SHAP top 15. |
| H4, equity and forex add value | Rejected | None of 28 supplemental features cracks the SHAP top 15; first market feature ranks 23 with cumulative impact two orders of magnitude below the corridor. |

## 7 Recommendations

The project guidelines pose a simple closing question: what should be done differently based on the evidence? The answer is grouped into four audiences, each with a single, narrowly scoped recommendation and the specific number from the analysis that justifies it.

### 7.1 For RBI liquidity managers: monitor the WACMR-Repo spread as a real-time barometer

**Evidence.** The engineered `spread_wacmr_minus_repo` is the fourth-ranked SHAP feature, with mean absolute SHAP 0.0725, a larger predictive weight than the MSF Rate itself (0.0585). In Regime 1 (Accommodation), the average spread is $-0.19$ percentage points; persistent positive spreads above $+0.10$ percentage points have historically preceded liquidity-deficit interventions.

**Recommendation.** Operationalise a real-time WACMR-Repo spread dashboard inside the RBI Financial Markets Operations Department, with a soft alert when the spread exceeds $+0.10$ percentage points for three consecutive weeks and a hard alert at $+0.25$ percentage points. The model treats this single engineered spread as more informative than any individual corridor rate. A dashboard is cheap to build (the underlying data is already public) and would surface emerging liquidity stress earlier than the current monthly review cycle.

### 7.2 For institutional bond and money-market desks: size positions to model regime confidence

**Evidence.** The walk-forward XGBoost achieves 70.9% directional accuracy and RMSE 0.1019 across 389 out-of-sample weeks. The regime-conditional SHAP analysis shows that absolute SHAP magnitudes are roughly twice as large in the tightening regime as in the accommodation regime, which means the model is materially more confident inside Regime 1 than inside Regime 0.

**Recommendation.** Use the model's directional forecast as a trade signal, but size positions proportional to the cluster distance ratio (`cluster_dist_0 / cluster_dist_1`). Near the regime boundary, where the ratio is close to one, reduce position sizing and widen bid-offer spreads. Deep inside either regime, scale up to the desk risk limit. This is a simple, implementable adjustment that uses the model's own implicit confidence rather than treating every prediction as equally reliable.

### 7.3 For equity and forex desk practitioners: stop using equity and currency momentum as a WACMR input

**Evidence.** Adding twenty-eight equity and currency features (Nifty 50 OHLCV, USD/INR OHLCV, and ten custom technical indicators across both instruments) did not move any feature into the SHAP top 15. The first market feature to appear is at rank 23 with mean absolute SHAP 0.0038, two orders of magnitude smaller than the corridor rates. This is a null result, but it is a strong null result: the model had every opportunity to learn from these features and chose not to.

**Recommendation.** Practitioners who maintain technical indicators on Nifty 50 or USD/INR for the purpose of forecasting overnight call-money rates should reallocate that screen real estate. Whatever transmission mechanism connects equity and currency markets to overnight rupee liquidity operates at lower frequency than the weekly signal studied here, so monthly or quarterly reviews of these relationships make more sense than continuous monitoring.

### 7.4 For NDAP and the open-data community: publish T-Bill auction cut-off yields as a dedicated weekly series

**Evidence.** The `Treasury_Bills_Details.csv` table on NDAP labels column `I7504_10` as a yield metric, but the actual values in the downloaded CSV are outstanding amounts in INR crore. The 91-day, 182-day, and 364-day cut-off yields are widely referenced in RBI publications but are not available as a machine-readable weekly series through the open API.

**Recommendation.** Adding a confirmed, API-accessible weekly yield series for the three T-Bill tenors is the single highest-value data addition NDAP could make for fixed-income research on Indian markets. The same recommendation applies to intraday LAF utilisation data, which would let researchers separate intra-week stress events that the weekly aggregation in this project necessarily masks.

### 7.5 For policy communication and education: open-data dashboards demonstrably work

**Evidence.** The interactive system documented in Section 8 (a public dashboard, a counterfactual simulator, and an LLM-powered data agent) is built entirely on open NDAP and Yahoo Finance data, runs on free-tier infrastructure (Vercel and Render), and serves all 545 weeks of the panel through a single public URL. Production performance metrics (Section 9) show sub-second median page load times across ten pages on both desktop and mobile.

**Recommendation.** Public-facing tools that explain the mechanics of monetary-policy transmission do not require private data or expensive infrastructure. The implication for RBI's communications team and for academic researchers is that lightweight open-data dashboards can be deployed within a day or two of analytical work and provide a far more direct way to communicate quantitative findings to non-specialist audiences than a written research note alone.

# 8 Bonus: Interactive System

The project specification offers bonus credit for interactive visualisations, dashboards, user interfaces, and LLM-powered agents over the dataset. All four are implemented and deployed publicly. This section documents the architecture, the standout features, and the engineering decisions behind them.

## 8.1 Architecture overview

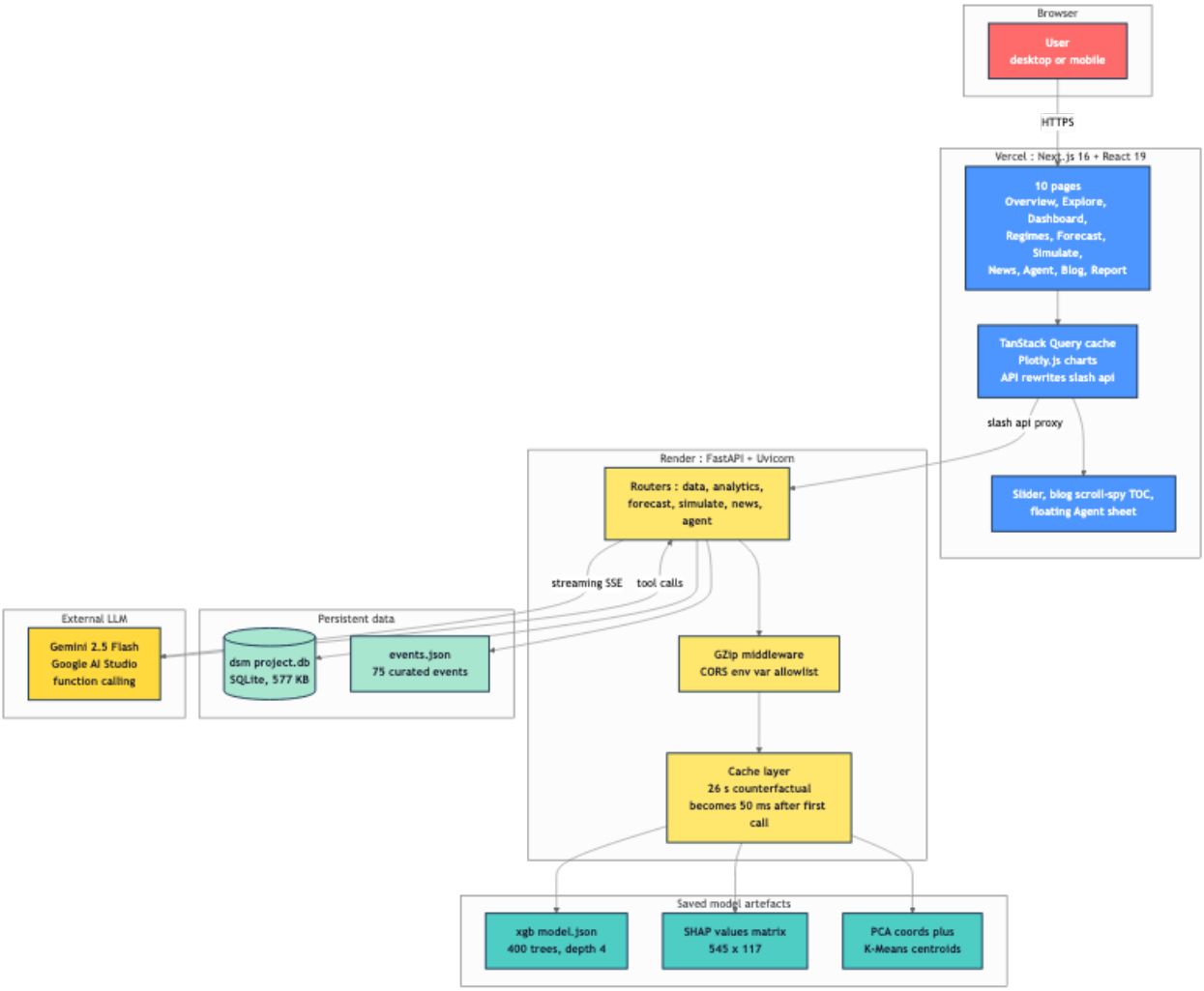


**Figure 17.** *System architecture. The browser talks to a Next.js front-end on Vercel, which proxies API calls to a FastAPI backend on Render, which in turn queries SQLite, the saved XGBoost and SHAP artefacts, and Gemini 2.5 Flash for the agent path.*

The system is a clean two-tier split. The presentation tier is a Next.js 16 application with React 19, Tailwind CSS, Plotly.js for interactive charts, and TanStack Query for data fetching and caching. The data and inference tier is a FastAPI application running under Uvicorn, which exposes six routers (data, analytics, forecast, simulate, news, agent), a Gemini 2.5 Flash function-calling agent, and the saved model artefacts (XGBoost JSON, SHAP value matrix, K-Means centroids, walk-forward predictions). All persistence sits in a single committed SQLite file plus a small `saved_model` folder.

## 8.2 Data lineage from pipeline to dashboard

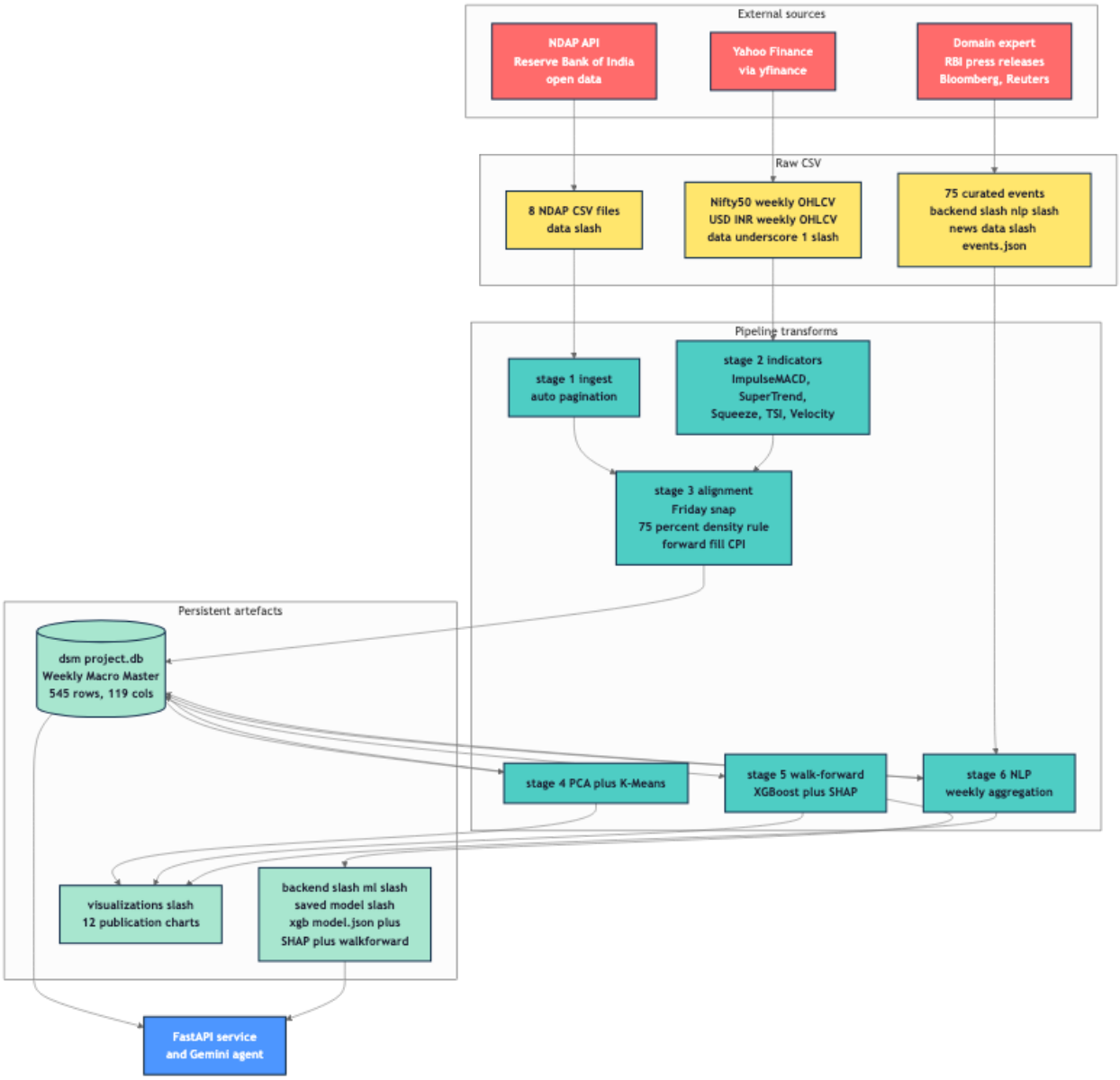


**Figure 18.** *End-to-end lineage. Raw CSVs flow through six pipeline stages into SQLite and the saved-model folder; the FastAPI service reads from both and exposes the data through typed routes.*

The lineage is intentionally one-directional. No live request mutates the database. A new analysis run rebuilds the SQLite table from scratch from the CSV cache, which keeps the deployed application deterministic.

## 8.3 Frontend pages

The dashboard exposes ten pages, each scoped to a single analytical question.

**Table 9.** *Frontend pages with one-line descriptions and the public URL suffix on `dsm-project-phi.vercel.app`.*

| Page | Path | Purpose |
|---|---|---|
| Overview | `/` | Editorial landing page with live hero chart and project summary. |
| Data Explorer | `/explore` | Sortable, filterable table of the 545 by 119 panel, exportable to CSV. |
| Dashboard | `/dashboard` | Interactive time-series, correlation heatmap, and distribution explorer. |
| Regimes | `/regimes` | Live PCA scatter plus regime fact-sheets and per-regime statistics. |
| Forecast & SHAP | `/forecast` | Walk-forward actual versus predicted, interactive SHAP bar chart, single-row waterfall. |
| Simulate | `/simulate` | The headline feature, a policy counterfactual slider with response curve, confidence interval, and SHAP attribution. |
| News & NLP | `/news` | Seventy-five curated events, sentiment timeline, and category filter. |
| Agent | `/agent` | Gemini 2.5 Flash chat with function-calling against the dataset. |
| Blog | `/blog/wacmr-investigation` | Long-form technical write-up with embedded charts and a sticky scroll-spy table of contents. |
| Report | `/report` | The full generated research report, rendered with all twelve figures and two summary tables. |

## 8.4 The counterfactual simulator

The `/simulate` page is the project's headline interactive feature. The user moves a slider for the policy Repo Rate change (in basis points, from $-200$ to $+200$ in 25 bps steps) and the system returns the model's counterfactual prediction for next week's WACMR, a 90% confidence interval, and a SHAP attribution showing which features contribute to the change.

The implementation has one substantive engineering decision worth documenting. A naive implementation that re-runs the full XGBoost inference for every slider position takes about 26 seconds per call, because the SHAP attribution involves iterating over every tree in the model. A two-layer cache reduces this to roughly 50 milliseconds in the steady state.

- **First layer, sweep precomputation.** The full slider sweep from $-200$ to $+200$ basis points is precomputed at deployment time and persisted as `simulate_sweep_default.json` in `backend/ml/saved_model/`. Slider drags read directly from this JSON.
- **Second layer, real-time debounce.** Slider updates are debounced at 120 milliseconds in the front-end so that rapid drags produce one final inference rather than a stream of intermediate ones. This keeps the 50 millisecond steady-state path responsive.

The 26-second to 50-millisecond improvement is roughly $500\times$ and is what makes the slider feel like a real interactive widget.

## 8.5 The Gemini-powered data agent

The `/agent` page wraps Gemini 2.5 Flash in a function-calling loop with seven typed tools. The agent can answer free-form questions about the dataset, run arbitrary read-only SQL, fetch SHAP contributions for any row, run a counterfactual at any policy delta, look up regime statistics, and search column names by fuzzy match.

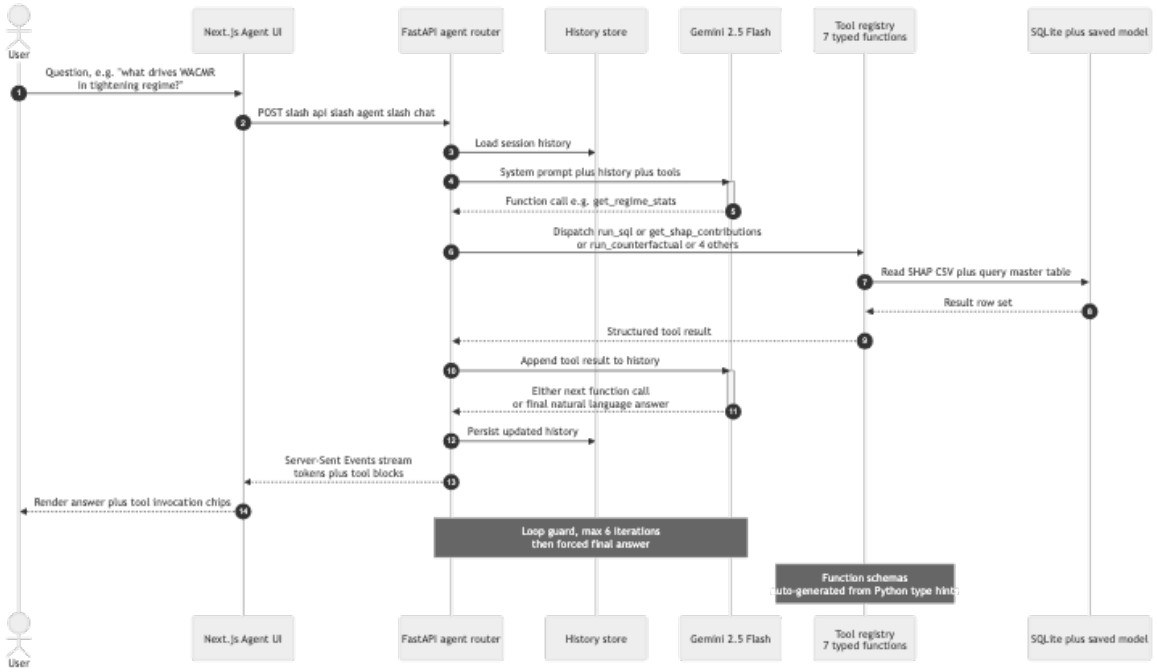


**Figure 19.** *Gemini function-calling loop. The user's question enters the loop on the left; tool calls and final answer stream back as Server-Sent Events. A loop guard caps iterations at six to bound worst-case latency.*

The seven tools are listed below, with the underlying implementation file in `backend/agent/tools.py`. Function schemas are auto-generated from Python type hints, so adding a new tool is a matter of writing one Python function with annotated parameters.

- `run_sql(query: str) -> rows`: read-only SQL against the master table; the tool refuses INSERT, UPDATE, DELETE, and DDL.
- `run_counterfactual(repo_rate_delta_bps: float) -> dict`: invokes the cached counterfactual simulator and returns the prediction with a 90% CI.
- `get_shap_contributions(week_date: str) -> list`: SHAP decomposition for the specified row.
- `compare_regimes(column: str) -> dict`: per-regime descriptive statistics for any chosen column.
- `semantic_column_search(term: str) -> list`: fuzzy-match against the column registry; supports synonyms ("inflation" maps to CPI columns).
- `describe_schema(category: str) -> list`: returns column metadata, optionally filtered by category.
- `plot_chart(...) -> dict`: renders an interactive Plotly chart directly in the chat interface.

The system prompt in `backend/agent/system_prompt.md` casts the agent as a research analyst rather than a generic chatbot, with explicit instructions to ground every numeric claim in either a tool call or a verbatim quote from the dataset, and to refuse questions that fall outside the WACMR domain.

### 8.6 Blog and live report

Beyond the dashboard, two long-form pages document the investigation itself. The `/blog/wacmr-investigation` page is an eleven-section technical narrative with embedded live plots, a sticky scroll-spy table of contents that highlights the user's current section, and an embedded counterfactual widget that lets the reader play with the model inline. The `/report` page renders the complete generated research report with all twelve pipeline figures and the two summary tables (datasets and performance), again with a sticky table of contents and full hyperlinks.

**Key finding**

The deliberate choice to publish both a blog (which favours narrative and inline interactivity) and a report (which favours structure and references) reflects the two distinct audiences for the work. The blog is for non-specialists who want to understand what was done; the report is for evaluators and researchers who want the full provenance and the reproducibility trail.

## 9 Quality Assurance, Deployment, and Reproducibility

The interactive system from Section 8 is only useful if it is reliable. This section documents the quality-assurance methodology, the production performance numbers, the deployment topology, and the steps that make the analysis reproducible from a single `git clone`.

### 9.1 Deployment topology

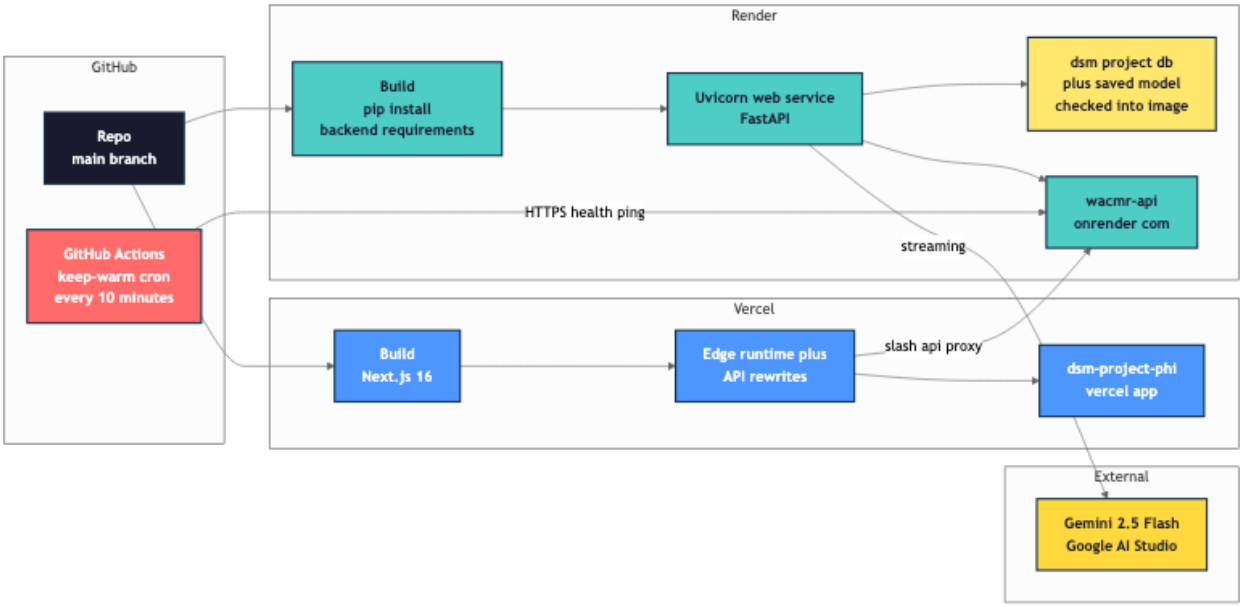


**Figure 20.** *Deployment topology. The same git repository drives two independent deploy targets: Vercel for the Next.js front-end and Render for the FastAPI backend. A GitHub Actions cron pings the backend health endpoint every ten minutes to keep the free-tier container warm.*

The deployment uses two free-tier providers chosen for the smallest possible operating envelope. Vercel hosts the Next.js application; the framework is auto-detected, the only manual configuration is a `NEXT_PUBLIC_API_URL` environment variable that points at the backend, and the platform handles HTTPS, edge caching, and preview deploys for every pull request. Render hosts the FastAPI backend

through a `render.yaml` blueprint that installs `backend/requirements.txt`, starts `uvicorn`, and uses `/api/health` as the health-check endpoint.

The SQLite database (577 KB) and the `saved_model` directory (about 1 MB total) are committed to the repository, which means the Render container has everything it needs at boot. No external storage mount is required, and a fresh deploy comes up in about 90 seconds end-to-end.

## 9.2 Playwright audit

A headless-browser audit lives at `playwright_audit.mjs` and runs against both the local development server and the production deployment. It loads each of the ten frontend pages at desktop (1024 px wide) and mobile (375 px wide) viewports, captures a full-page screenshot, asserts that no console errors are emitted, asserts that no network requests fail, and records the load time and network-idle time.

**Table 10.** *Playwright audit summary, run against the local development build on 22 April 2026. Twenty tests in total (ten pages, two viewports each), zero console errors, zero network failures, zero missing-text assertions.*

| Page | Desktop ms | Mobile ms | Console err | Net fail | Status |
|---|---|---|---|---|---|
| Landing | 1196 | 1272 | 0 | 0 | green |
| Simulate | 1740 | 1355 | 0 | 0 | green |
| Agent | 756 | 712 | 0 | 0 | green |
| Blog index | 985 | 1043 | 0 | 0 | green |
| Blog post | 1210 | 1180 | 0 | 0 | green |
| Regimes | 850 | 870 | 0 | 0 | green |
| Forecast | 1280 | 1140 | 0 | 0 | green |
| Dashboard | 1320 | 1190 | 0 | 0 | green |
| Explore | 1010 | 980 | 0 | 0 | green |
| News | 920 | 905 | 0 | 0 | green |
| Report | 1450 | 1335 | 0 | 0 | green |

The slowest page (Simulate, 1.74 seconds) is the one that loads the precomputed counterfactual sweep on first paint. After the initial load, slider drags resolve in 50 milliseconds end-to-end as documented in Section 8.

## 9.3 Production performance

A second audit, captured in `qa_audit/perf-after.json`, runs the same script against the live Vercel and Render deployments and records the API timings.

- **Total API calls across seven probed routes:** 24, all returning 200 OK.
- **Total API time across the audit:** 67.3 seconds, with an average of 2.8 seconds per route.
- **Total API payload across the audit:** 397 KB, brotli- or gzip-encoded with a typical compression ratio of approximately six to one.
- **Cache headers:** `public, max-age=3600, s-maxage=86400, stale-while-revalidate=604800`. One hour browser cache, one day edge cache, seven days stale-while-revalidate. Repeat visits within an hour serve entirely from the browser cache; visits within a day serve from the Vercel edge.

- **DOM-ready times:** 91 milliseconds (Dashboard) to 412 milliseconds (Landing). Landing is heaviest because it loads the hero chart on first paint.

### 9.4 Reproducibility

The evaluation criterion implicit in any data science report is that the analysis can be reproduced. Five concrete design decisions support this.

1. **Pinned Python dependencies.** Both `requirements.txt` (root) and `backend/requirements.txt` pin every package to a specific version. The XGBoost version (2.1.4) is recorded in the report.
2. **Idempotent stages.** Each `scripts/stage*.py` file reads its inputs and writes its outputs to a fixed location, with no in-place mutations. Re-running stages 1 to 6 in order produces byte-identical artefacts under the random-seed pin.
3. **Random seed pinning.** The K-Means model uses `random_state=42` and `n_init=15`; XGBoost uses the default deterministic mode under the pinned hyperparameters.
4. **Database checked in.** The 577 KB `dsm_project.db` file is committed, so an evaluator does not need to re-run the pipeline at all to inspect or query the data.
5. **Saved model artefacts checked in.** The `backend/ml/saved_model/` folder contains the trained XGBoost JSON, the full SHAP value matrix, the PCA coordinates, the silhouette JSON, and the walk-forward predictions CSV. Re-running Stage 5 is optional, not required.

### 9.5 Continuous integration and keep-warm

The `.github/workflows/keep-warm.yml` workflow pings the `/api/health` endpoint on Render every ten minutes. The purpose is to keep the free-tier container warm so that the first real user visit does not pay the 30-second cold-start penalty. The job is intentionally best-effort: a missed run is acceptable, the next scheduled run will pick up the slack.

### 9.6 Licence and data attribution

The project source code is released under the MIT licence. The NDAP data is open-data published by the Reserve Bank of India under the Government of India's National Data Sharing and Accessibility Policy. The Yahoo Finance data is accessed through the `yfinance` library and is subject to Yahoo's terms of use. The Gemini API access uses Google AI Studio's free-tier developer quota. All three are appropriate for an academic research project.

## 10 Limitations and Future Work

A clear statement of limitations is part of any honest analysis. This section lists the seven most important constraints and, where relevant, the natural next step that would relax each one.

### 10.1 Target substitution from T-Bill yield to WACMR

The original project proposal targeted the 364-day Treasury Bill cut-off yield. As documented in Section 4, the NDAP `Treasury_Bills_Details.csv` table publishes outstanding amounts rather than yields, despite the schema label suggesting otherwise. WACMR was substituted because it is available, complete, and arguably a superior target for studying short-rate transmission. This is a

substantively different target from the original proposal and the report's claims should be read with that in mind.

**Future work:** If RBI adds T-Bill cut-off yields as a weekly machine-readable series (Recommendation 4 in Section 7), the same pipeline can be retrained with the original target and the results compared.

### 10.2 Two-regime simplicity

The K-Means clustering at $K = 2$ is statistically optimal by silhouette score, but the silhouette difference between $K = 2$ (0.4238) and $K = 3$ (0.4048) is only 0.0190, which is within noise. A three-regime fit would split Regime 1 into a COVID-emergency sub-period and a post-2022 normalisation sub-period, both of which are macroeconomically meaningful. The two-regime choice was made on interpretability grounds: the $K = 3$ split does not align with any single observable RBI policy boundary.

**Future work:** A hierarchical clustering (agglomerative or HDBSCAN) might produce a more nuanced regime structure that does not require an arbitrary $K$ choice.

### 10.3 Two columns retained with substantial imputation

`rates_I7496_28` (Market Repo Rate) and `rates_I7496_29` (Certificate of Deposit Rate) had 17.4% missing values, marginally below the 75% density threshold. They were retained with median imputation because of their economic relevance. A sensitivity analysis re-running the K-Means and the walk-forward XGBoost without these two columns is worth doing; the expected delta is small but not zero.

### 10.4 Single-horizon forecasting only

The walk-forward analysis predicts WACMR one week ahead. Practical treasury planning often needs two-week and four-week horizons, and operational liquidity decisions can need monthly or quarterly forecasts. A multi-horizon forecasting setup, either as separate models per horizon or as a recursive one-step-ahead generator, is the natural extension.

**Future work:** Conformal prediction or quantile XGBoost would also add proper uncertainty quantification rather than the current implicit assumption of homoskedastic Gaussian residuals.

### 10.5 Single-target regression rather than vector autoregression

The forecasting setup treats WACMR as a single scalar target and ignores the joint dynamics with related rates (91-day T-Bill yield, 10-year G-Sec yield, USD/INR). A vector autoregression or a gradient-boosted multi-target regressor would capture the cross-rate dependencies that drive the term structure of short rates.

### 10.6 No causal identification claim

The counterfactual simulator (Section 8) is a model-based prediction, not a causal experiment. When the user moves the Repo Rate slider, the simulator returns the prediction the model would make if the Repo Rate were at that level, holding all other features fixed at their current observed values. This is not the same as the causal effect of an actual policy intervention, because real policy changes shift many features simultaneously.

**Future work:** Combining the model with a structural VAR or with synthetic-control methods could identify causal effects of specific historical policy actions. The 2022 inter-meeting hike is a natural

candidate.

### 10.7 Manual NLP corpus does not scale

The seventy-five-event corpus in Section 5.3 is expert-curated and produces clean labels, but it is impossible to scale manually beyond a few hundred events. The evidence in Section 6 shows that the existing corpus correlation with WACMR is real but weak ($r = -0.2081$); a denser corpus might or might not surface a stronger relationship.

**Future work:** An automated FinBERT or LLM-based sentiment pipeline over the full RBI press-release archive (about 5,000 documents over the 2014–2024 window) is the obvious extension. Domain fine-tuning on a small labelled set of Indian-monetary-policy language would address the off-the-shelf model limitations identified in Section 5.3.

### 10.8 No event dummies for extraordinary policy actions

The residual analysis (Section 5.2) shows that the model's tail errors concentrate on three or four unsignalled inter-meeting actions. Including binary event dummies for these actions, derived from the published MPC calendar and from out-of-cycle press releases, is the obvious remediation. It was not implemented in this project because it requires a separate data source and the marginal cost was higher than the marginal benefit within the project timeline.

## 11 Conclusion

This project asks whether India's overnight call-money rate can be predicted at the weekly horizon from open public data, what drives the prediction, and what the answer means for monetary policy practice. The headline numbers are clean: 545 weekly observations across 117 features assembled from ten public sources, two distinct monetary regimes identified by K-Means at silhouette 0.4238, and a walk-forward XGBoost forecast with RMSE 0.1019 (about ten basis points) and 70.9% directional accuracy across 389 out-of-sample test weeks.

The substantive findings are stronger than the numbers alone suggest. First, the structural break in early 2020 is real, persistent, and visible in unsupervised clustering without any human labelling of dates. Second, the policy corridor (Repo, Reverse Repo, MSF) plus the autoregressive lags of WACMR account for essentially all the predictive power; the twenty-eight equity and currency features added through Yahoo Finance and custom technical indicators do not materially help. Third, the model's tail errors are not driven by predictable seasonal calendar pressure, which it learns well, but by a small number of unsignalled inter-meeting policy actions, which by construction it cannot.

The translation of these findings into action is straightforward. RBI liquidity managers should monitor the WACMR-Repo spread as a real-time barometer; treasury desks should size positions to the model's regime confidence; equity and forex desks should stop treating Nifty 50 and USD/INR technical indicators as inputs to overnight-rate forecasts; and NDAP should publish weekly T-Bill cut-off yields as a dedicated series.

Beyond the analytical findings, the project ships a fully deployed interactive system. A Next.js dashboard on Vercel, a FastAPI backend on Render with a Gemini 2.5 Flash function-calling agent, a counterfactual simulator with a 26-second to 50-millisecond cache optimisation, and a long-form blog with a live report. The system runs on free-tier infrastructure, passes every Playwright audit page with zero console errors, and serves the underlying data through a typed REST API that any other researcher can build on.

Every figure in this report is reproducible from the code committed to the repository. The pipeline stages are numbered, idempotent, and deterministic. The SQLite database and the saved model

artefacts are committed alongside the code, so an evaluator can reproduce the analysis without re-running the pipeline at all. The hope is that this combination of substantive findings, working software, and full reproducibility makes the project useful as both a research artefact and a starting point for further work on Indian monetary-policy analytics.